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GMM estimation with cross sectional dependence

T.G. Conley

Department of Economics, Northwestern University, Evanston, IL 60208, USA

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Abstract

This paper presents a spatial model of dependence among agents using a metric of *economic distance*. Measurements of this economic distance provide cross-sectional data with a structure similar to that provided by the time index in time-series data. Generalized method of moments estimators using such dependent data are shown to be consistent and asymptotically normal. This paper presents a class of non-parametric, positive semi-definite covariance matrix estimators that allow for general forms of dependence characterized by economic distance. These covariance matrix estimators are shown to remain consistent when economic distances are not precisely observed. © 1999 Elsevier Science S.A. All rights reserved.

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1. Introduction

Economic theory predicts that agents take actions that lead to interdependence among themselves. For example, the prediction that risk averse agents will make insurance contracts allowing them to smooth idiosyncratic shocks implies dependence in consumption across individuals. Trade can indirectly induce interdependence in agents' activities as well. Real shocks, even to remote areas, can have an impact, via price changes, on the rest of the economy. From the econometrician's point of view, these interdependencies will result in dependence among individuals' unobservables. Thus the independence assumptions prevalent in cross-sectional econometrics are often at odds with economic theory.

However, in econometrics, independence assumptions are made to facilitate estimation and inference and may not accurately reflect reality. If an independence assumption is to be disposed of, it must be replaced with assumptions on the nature of the dependence between observations. Approaches in the literature usually involve strong parametric assumptions on the distributions of the data and error terms, as in, e.g., Moulton (1990) and Case (1991). General forms of dependence are rarely allowed for in cross-sectional data, although routinely allowed for in time-series data. A major reason for this is that a typical time series comes with a time label giving the data a natural ordering and structure absent in cross-sectional data. In practice, this makes estimation and inference given minimal assumptions about dependence more difficult with cross-sectional data than with time-series data.

This paper presents a model that adds structure to cross-sectional data by using other information available to the econometrician. Specifically, we argue that a spatial model of dependence between observations can be useful to model dependence among economic agents. Each agent's observation is modelled as a realization of a random process at a point in a Euclidean space: a random field.¹ The distance between two agents in this space reflects their proximity with respect to unobservables, a notion of 'economic distance'. The joint distribution of random variables at a set of points is a function of the economic distances between them. This modelling strategy allows for a simple characterization of considerable interdependence among agents. It allows more complicated dependence across individuals than models with group-specific effects, e.g. Moulton (1990), or with scalar indexed dependence, e.g. White and Domowitz (1984). However, it requires that the econometrician have information regarding this economic distance.

This approach will be useful for applications in urban, environmental, development, growth and other areas of economics where this kind of information is available. Economic distance information will differ greatly with the application. Applications in urban economics include investigations of peoples' willingness to pay for amenities.² Unobserved local public goods may induce correlations in observed prices. If this occurs, the relevant economic distance may be the time and monetary cost of travelling between points to use these local public goods. If however, the amenity to be priced is air quality, a topic relevant in environmental economics, then local weather conditions might constitute the major unobservables. Weather is obviously correlated at physically nearby locations so the best measure of economic distance in such applications is likely to be their physical distance. In an application like that in Case (1991), involving estimation

¹ Other applications of random fields in econometrics include Case (1991), Quah (1992), Elliott (1993), Driscoll and Kraay (1995), and Kelejian and Prucha (1995).

² For example, Roback (1982).

of Indonesian rice demand functions, physical distance may again be the best proxy of economic distance due to the nature of individual-specific transaction costs. Other applications in development economics include studies of risk sharing in rural developing economies.³ The primary shocks to individuals in such agrarian economies may be weather related. If so, measures of weather correlation on farms of two individuals could proxy for the economic distance between them. A large portion of the economic growth literature looks at data across countries to investigate correlations between GNP growth rates and a host of other variables.⁴ Even the most casual observation suggests that some countries, e.g., the U.S. and Canada, will have correlated unobservables. There are many potential measures of the economic distance between two countries, such as trade volumes and transportation costs of physical or human capital. Each of the measures above would have its merits in describing an aspect of economic distance in the associated application, yet most are imperfect measures. Thus a strategy for allowing for imperfectly measured economic distances is needed.

Imperfect distance measurements create problems for parametric models of spatial dependence. The typical approach to modelling spatial dependence is to assume the data is Gaussian with a covariance structure that is a function of known distances. Often it is assumed that a spatial analog to an auto-regressive moving-average (ARMA) model captures the dependence across individuals as in, e.g., Case (1991) or Anselin and Griffith (1988). When economic distances are measured with error, the distribution of measurement errors must be assumed known or likelihood based estimators will be misspecified. As is apparent in the above examples, economic theory provides guidance as to potential measurements of economic distance but not about specific measurement error distributions. The econometrician will likely be confident that his economic distance measurements are in the right ball park – measurement errors are bounded – but unsure of a good error distribution assumption. In order to avoid such assumptions and the potentially complicated likelihood-based estimation they would require, we focus on moment-based estimation and take a nonparametric approach to modelling dependence.

This paper addresses the complications that a spatial model of dependent data presents for a Generalized Method of Moments (GMM) estimator (Hansen, 1982). GMM estimators remain consistent with such dependent data but their distribution theory, covariance matrix estimation procedure, and the efficient GMM estimator are different with this spatial model of dependence than with a time series model of dependence. The approach followed here is to

³ For example, Townsend (1994).

⁴ For example, Barro (1991).

estimate covariance matrices nonparametrically, allowing for spatial dependence. The time-series analog of this approach is roughly that followed by, e.g., Newey and West (1987), Andrews (1991), White (1984), or White and Domowitz (1984).

Covariance matrix estimation is discussed in two situations, the first is when the economic distances between observations are measured exactly. The second is when only inexact distance measurements are observed; true distances are distorted by bounded measurement errors. In the first case, a class of positive semi-definite (PSD) estimators of asymptotic covariance matrices is presented. In the second case, these covariance matrix estimators are shown to remain consistent when it is assumed that true distances are distorted by bounded, mixing measurement error. We present a method to construct a PSD estimator when measurement errors satisfy an additional condition.

The remainder of this paper is organized as follows. Section 2 provides an overview of the basic econometric model. The next two sections present large sample results for GMM estimators and covariance matrix estimators using spatially dependent data. Section 3 presents a simple model of spatial dependence, where agents live on an integer lattice and economic distances are known. Section 4 presents a more general model that allows agents to live at real-valued locations and measurements of distances to be distorted by bounded errors. Section 5 contains a brief empirical illustration of the potential impact of using the paper's econometric model. Finally, Section 6 concludes with a discussion of limitations and possible extensions.

2. Basic model

The population of individuals is assumed to reside in a Euclidean space, taken for the sake of exposition to be $\mathbb{R}^2$, with each individual i located at a point s_i . The econometrician's sample consists of realizations of agents' random variables at a collection of locations $\{s_i\}$ inside a sample region. The sample region is one of a sequence of finite closed, convex regions $\{A_\tau\}$ which increases in area as $\tau \rightarrow \infty$. The population of potentially observable (as the region A_τ grows) locations forms a lattice with (potentially) irregular spacing denoted H . As illustrated in Fig. 1 the sample consists of $H \cap A_\tau$. A_τ is assumed to grow uniformly in at least two non-opposing directions, to increase the sample size as $\tau \rightarrow \infty$.⁵ Associated with each position s is a vector of random variables X_s called a random field.

⁵ This assumption is made to ensure that vector indexing of dependence is not superfluous. If the A_τ remained between two parallel lines, then dependence could be scalar-indexed, it could be handled like the data were a time series.

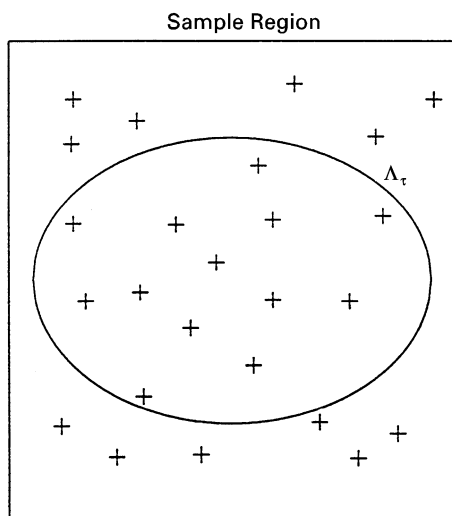


Fig. 1. The econometrician's sample consists of variables observed at points within A_τ and measurements of interpoint distances.

In a given sample, the econometrician's data consists of two parts, the first is a realization of the random field X_{s_i} at all points s_i within the set A_τ , i.e. $\{X_{s_i}: s_i \in A_\tau\}$. The number of points in A_τ is a random variable T_τ . The second part of the data is a $T_\tau \times T_\tau$ symmetric matrix D whose elements d_{ij} are measurements of the pair-wise distances between the sample indexes of points in A_τ .

We assume that economic theory has produced a moment condition involving X_{s_i} that can be used to estimate a parameter vector β that is of interest. The goal of this paper is to provide conditions for the consistency and asymptotic normality of the GMM estimator of β and to provide tractable PSD covariance matrix estimators when the X_{s_i} are dependent.

The basic model of dependence is that the distance between agents' positions, corresponding to their economic distances, characterizes the dependence between their random fields. If two agents' locations s_i and s_j are close then their X_{s_i} and X_{s_j} may be very highly correlated. As the distance between s_i and s_j grows large, the random variables X_{s_i} and X_{s_j} become closer to being independent in a sense made precise below.

3. Estimation with known economic distances

3.1. Data generation

In this section the population is assumed to reside at integer coordinate locations in order to simplify the presentation of the econometric model with known distances. Since economic distance measurements are exact, the positions s_i of individuals in the sample can be inferred up to a normalization of location and orientation from the interpoint distances. A method of finding coordinates of points in a Euclidean space given their interpoint distances is available in, e.g., Mardia et al. (1979).⁶ Because the econometric theory here does not depend on these two normalizations, locations are taken to be observed in this section.

The data generating process has two components. The first part is a sampling process that determines which individuals and hence which locations $\{s_i\}$ are observed. The values of attributes of individual i are determined by the value of an underlying random field X_s at location s_i . Following Clark (1973), the observable random field on the lattice H , X_{s_i} , is said to be subordinated to the random field X_s and the process generating H is called the directing process.

3.1.1. Underlying random field

The value of random variables at each location s_i is determined by a $\ell \times 1$ random field X_s with the index $s \in \mathbb{Z}^2$. The X_s are assumed to be stationary, i.e. the joint distribution of X_s for any collection of indices s is invariant to a shift in the indices. X_s is also assumed to be mixing.

3.1.2. Directing process

The directing process determining H can be described by a regular lattice indexed random field W_s that is equal to one if location s is sampled and zero otherwise. The W_s process is assumed to be independent of the underlying random field. The W_s process is also assumed to have a finite expectation $EW_s = \lambda$, and to be stationary and mixing.

The directing process W_s can be viewed as arising from both the distribution of the population on the lattice and the actual survey sampling scheme used in collecting the data. For example, if the population of agents were evenly distributed on the plane and the sample was an independent sample of agents then W_s would be an IID dummy variable. If, however, the population was concentrated in certain parts of the plane and agents were sampled independently then W_s would exhibit positive spatial correlation. However, spatial

⁶ This is a special case of a method, called multidimensional scaling, to fit a configuration of points in $\mathbb{R}^k$ whose interpoint distances approximate an arbitrary (perhaps non-Euclidean) distance matrix.

correlation in W_s could also arise from cluster sampling of an evenly distributed population. Thus it is not possible to distinguish sampling schemes from population distributions given locations alone.

3.1.3. Mixing coefficient

Let $\mathfrak{I}_A$ be the σ -algebra generated by a given random field ψ_{s_m} , $s_m \in A$ with A compact, and let $|A|$ be the number of $s_m \in A$. Let $Y(A_1, A_2)$ denote the minimum Euclidean distance from an element of A_1 to an element of A_2 . The mixing coefficient used is defined as

$$\alpha_{k,l}(n) \equiv \sup\{|P(A \cap B) - P(A)P(B)|\}, A \in \mathfrak{I}_{A_1}, B \in \mathfrak{I}_{A_2} \quad \text{and} \\ |A_1| \leq k, |A_2| \leq l, Y(A_1, A_2) \geq n \quad (3.1)$$

Our definition of mixing for X_s and W_s requires $\alpha_{k,l}(n)$ for each process to converge to zero as $n \rightarrow \infty$ at a rate which will be specified below.

3.2. Estimation problem

Economic theory has produced a moment restriction that can be used to get estimates of a parameter β that is identified as the unique solution to the following moment equation:

$$\text{Eg}(X_{s,v}; \beta) = 0, \quad (3.2)$$

where $g: \mathfrak{R}^k \times \mathcal{B} \rightarrow \mathfrak{R}^v$. β is a $k \times 1$ vector of parameters to be estimated and β is in the interior of $\mathcal{B}$, a compact subset of $\mathfrak{R}^k$. v is assumed greater than or equal to k so there are $v - k$ overidentifying restrictions. The GMM estimator of β , b_τ , is the argmin of

$$J(b)_\tau \equiv \left[\frac{1}{T_\tau} \sum_{i=1}^{T_\tau} g(X_{s_i}; b) \right]' \Omega_\tau \left[\frac{1}{T_\tau} \sum_{i=1}^{T_\tau} g(X_{s_i}; b) \right] \quad (3.3)$$

over all b in $\mathcal{B}$, where Ω_τ is a positive-definite weighting matrix. If Ω_τ is a consistent estimator of the asymptotic covariance matrix of the moment conditions, we obtain efficient GMM estimates of β (Hansen, 1982).

3.2.1. Asymptotic properties of b_τ

Consistency: A set of sufficient conditions for consistency is analogous to the set of conditions assumed by Hansen (1982) in the time-series case. Essentially, the only difference between this spatial model and the time-series case is that

a pointwise law of large numbers for random fields rather than one for time series is used.

A1. A_τ grows uniformly in two non-opposing directions as $\tau \rightarrow \infty$.

A2. $\Omega_\tau \rightarrow \Omega$ in probability where Ω is a positive-definite matrix.

A3. X_s and W_s are mixing; $g(\cdot; b)$ is Borel measurable for all $b \in B$; and $g(X; \cdot)$ is continuous on B for all $x \in \mathfrak{R}^\ell$, and first moment continuous on B .⁷

Proposition 1. Given conditions A1–A3, $b_\tau \rightarrow \beta$ in probability as $\tau \rightarrow \infty$.

Proof. See the appendix.

Asymptotic distribution: The first-order conditions from minimizing Eq. (3.3) are

$$\left[\frac{1}{T_\tau} \sum_{i=1}^{T_\tau} Dg(x_{s_i}; b_\tau) \right]' \Omega_\tau \left[\frac{1}{T_\tau} \sum_{i=1}^{T_\tau} g(X_{s_i}; b_\tau) \right] = 0 \quad (3.4)$$

where $Dg(\cdot; \cdot)$ denotes the derivative of g with respect to b . Taking the usual mean value expansion of $g(X_{s_i}; b_\tau)$ about $\bar{b}$ yields

$$g(X_{s_i}; b_\tau) = g(X_{s_i}; \beta) + Dg(X_{s_i}; \bar{b})(b_\tau - \beta), \quad (3.5)$$

where $\bar{b}$ has elements between b_τ and β . Combining Eqs. (3.4) and (3.5) yields the familiar expression for $(b_\tau - \beta)$:

$$\begin{aligned} \sqrt{T_\tau}(b_\tau - \beta) = & - \left\{ \left[\frac{1}{T_\tau} \sum_{i=1}^{T_\tau} Dg(X_{s_i}; b_\tau) \right]' \Omega_\tau \left[\frac{1}{T_\tau} \sum_{i=1}^{T_\tau} Dg(X_{s_i}; \bar{b}) \right] \right\}^{-1} \\ & \times \left[\frac{1}{T_\tau} \sum_{i=1}^{T_\tau} Dg(X_{s_i}; b_\tau) \right]' \Omega_\tau \frac{1}{\sqrt{T_\tau}} \sum_{i=1}^{T_\tau} g(X_{s_i}; \beta) \end{aligned} \quad (3.6)$$

The conditions for consistent estimation of the portion of the expression in Eq. (3.6) that converges in probability,

$$\left\{ \left[\frac{1}{T_\tau} \sum_{i=1}^{T_\tau} Dg(X_{s_i}; b_\tau) \right]' \Omega_\tau \left[\frac{1}{T_\tau} \sum_{i=1}^{T_\tau} Dg(X_{s_i}; \bar{b}) \right] \right\}^{-1} \left[\frac{1}{T_\tau} \sum_{i=1}^{T_\tau} Dg(X_{s_i}; b_\tau) \right]' \Omega_\tau \quad (3.7)$$

⁷ The definition of first moment continuity for a random function $g(X_s; b)$ at a point b_0 is that defining $\text{mod}_g(\delta, b) = \sup\{|g(X_s; b) - g(X_s; \tilde{b})| : \tilde{b} \in B \text{ and } \|b - \tilde{b}\| < \delta\}$, $\lim_{\delta \rightarrow 0} E \text{mod}_g(\delta, b) = 0$ at $b = b_0$.

are essentially the same as in the time-series case.⁸

Regularity conditions for the portion of the expression in Eq. (3.6) that has an asymptotic distribution:

$$\frac{1}{\sqrt{T_\tau}} \sum_{i=1:T_\tau} g(X_{s_i}; \beta) \quad (3.8)$$

are slightly different from the time-series case. A central limit result is obtained by utilizing a central limit theorem due to Bolthausen (1982) for stationary, mixing random fields on regular lattices.

The expressions for limit theorems below are simplified by defining a process that is observed at all points within the sample region A_τ so that there are no missing observations. Let a new process $Y_s(b)$ take on the value at a point s of

$$Y_s(b) \equiv \begin{cases} g(X_s; b) & \text{if } W_s = 1, \\ 0 & \text{if } W_s = 0. \end{cases} \quad (3.9)$$

Obviously partial sums of $Y_s(b)$ equal partial sums of $g(X_{s_i}; \beta)$. Now express $(1/\sqrt{T_\tau}) \sum_{i=1}^{T_\tau} g(X_{s_i}; \beta)$ in terms of $Y_s(\beta)$:

$$\frac{1}{\sqrt{T_\tau}} \sum_{i=1}^{T_\tau} g(X_{s_i}; \beta) = \frac{\sqrt{|A_\tau|}}{\sqrt{T_\tau}} \frac{1}{\sqrt{|A_\tau|}} \sum_{s \in A_\tau} Y_s(\beta), \quad (3.10)$$

where the notation $|A_\tau|$ refers to the number of integer lattice locations within A_τ . The strategy for obtaining a limit distribution for Eq. (3.8) is to apply Bolthausen's central limit theorem to $(1/\sqrt{|A_\tau|}) \sum_{s \in A_\tau} Y_s(\beta)$ and show that $\sqrt{|A_\tau|}/\sqrt{T_\tau}$ converges in probability.

Additional moment and mixing conditions are required to get a limiting distribution for b_τ , a sufficient set is as follows. X_s and W_s are assumed to satisfy the following mixing and moment conditions B1–B3. Conditions B4 and B5 ensure that the expected derivatives are consistently estimated and expectations of derivatives and covariance matrices have full rank:

B1. $\sum_{m=1}^{\infty} m \alpha_{k,l}(m) < \infty$ for $k + l \leq 4$.

B2. $\alpha_{1,\infty}(m) = o(m^{-2})$.

B3. For some $\delta > 0$, $E(\|g(X_s; \beta)\|)^{2+\delta} < \infty$ and $\sum_{m=1}^{\infty} m \alpha_{1,1}(m)^{\delta/(2+\delta)} < \infty$.

⁸ The assumption here that the moment function $g(X_s; b)$ be differentiable is not necessary. For example, if $g(X_s; b)$ were assumed to satisfy a Lipschitz condition at β with an integrable Lipschitz coefficient then the strategy exposited in Hansen et al. (1995) or Newey and McFadden (1994) could be used to obtain an asymptotic normality result. We retain this differentiability assumption for pedagogical simplicity.

B4. $Dg(X_{s_i}; b)$ is Borel measurable for all $b \in B$, continuous on B for all $X \in \mathfrak{R}'$, and first moment continuous; $EDg(X_{s_i}; \beta)$ exists and is full rank;

B5. Defining $V = \sum_{s \in \mathbb{Z}^2} \text{Cov}(Y_0(\beta), Y_s(\beta))$, V is non-singular matrix.

Proposition 2. If conditions A1–A3 and B1–B5 are satisfied then, letting $\Rightarrow$ denote convergence in distribution,

$$\sqrt{T_\tau}(b_\tau - \beta) \Rightarrow N(0, D'_0 \lambda^{-1} V D_0) \quad \text{as } \tau \rightarrow \infty,$$

where

$$D'_0 = \{EDg(X_{s_i}; \beta)' \Omega EDg(X_{s_i}; \beta)\}^{-1} EDg(X_{s_i}; \beta)' \Omega. \quad (3.11)$$

Proof. See the appendix.

3.3. Covariance matrix estimation

The asymptotic covariance matrix of the moment conditions that needs to be estimated is

$$C \equiv \lambda^{-1} V = \lambda^{-1} \sum_{s \in \mathbb{Z}^2} E Y_0(\beta) Y_s(\beta)'. \quad (3.12)$$

For this section it is convenient to refer to the two coordinates in s directly so let $s = [m, n]$; also take the region A_τ to be a rectangle so that $m \in \{1, 2, \dots, M\}$ and $n \in \{1, 2, \dots, N\}$ where the dependence of M and N on τ is suppressed to save notation.⁹

The expression in Eq. (3.12) can be interpreted as a sum of spatial autocovariances, analogous to the time-series case where the infinite sum of an autocovariance function is the asymptotic covariance matrix of sample averages. A spectral representation of covariance stationary processes on the plane is also available, allowing the interpretation of the covariance matrix in Eq. (3.12) as a spectral density at frequency zero.

Familiar discrete time-series spectral density estimation techniques generalize to random fields on integer lattices.¹⁰ Smoothed periodograms can be used to estimate two-dimensional spectra (see, e.g., Priestley, 1982). For the special cases

⁹ An extension to non-rectangular A_τ is straightforward. The only role this particular shape sample region plays is to make indexing of points easy. The following proofs will go through for any sequence of regions satisfying the boundary conditions with very tedious notational modifications.

¹⁰ See Yaglom (1987) for an extensive discussion of spectral representations of random fields.

where $Y_{m,n}(\beta)$ is Gaussian or is a linear process, smoothed periodogram spectral density estimators have been proven to be consistent by Grenander and Rosenblatt (1957). However, the above restrictions on X_s , g , and W_s do not imply that $Y_{m,n}(\beta)$ is Gaussian or linear. Furthermore, the fact that the parameter β is unknown and to be replaced by the estimate b_τ must be addressed.

The class of covariance matrix estimators we consider is that formed by taking weighted averages of spatial autocovariance terms with weights that are zero for points farther than L_M and L_N apart in each direction. This class is analogous to time-series spectral density estimators whose time domain weights equal zero after a cutoff lag. The Bartlett window estimator used by Newey and West (1987) and the truncated estimator in White (1984) are two such time-series covariance matrix estimators.

Consider the following estimator of V , constructed as a weighted average of products of $Y_{m,n}$ terms:

$$\hat{V} \equiv \frac{1}{MN} \sum_{j=0}^{L_M} \sum_{k=0}^{L_N} \sum_{m=j+1}^M \sum_{n=k+1}^N K_{MN}(j, k) \left(Y_{m,n}(\beta) Y_{m-j, n-k}(\beta)' + \right. \\ \left. - \frac{1}{MN} \sum_{m=1}^M \sum_{n=1}^N Y_{m,n}(\beta) Y_{m,n}(\beta)' \right)$$

For an array of weights $K_{MN}(j, k)$ that are uniformly bounded and are such that $K_{MN}(0, 0) = 1$ and $K_{MN}(j, k) \rightarrow 1$ as $M \rightarrow \infty$ and $N \rightarrow \infty$. The subtraction of the second term is simply to remedy double counting of the $j = 0$ and $k = 0$ term in the first sum. With known economic distances such an array is simple to construct, a truncated weighting array with weights equal to one for $j < L_M$, $k < L_N$ is one example.¹¹ As detailed below, a careful choice of weighting arrays will produce covariance matrix estimates that are necessarily PSD.

If β were known, then we could use $(T_\tau/|A_\tau|)^{-1} = (T_\tau/MN)^{-1}$ to estimate λ^{-1} and estimate C by $(T_\tau/MN)^{-1} \hat{V}$. This estimator is of course not feasible because $Y_{m,n}(\beta)$ is not observed as β is unknown. The simple solution is to plug

¹¹ Precise distance information is not needed to obtain a consistent covariance matrix estimate. If distance information were only known up to broad categories then a weighting array $K_{MN}(j, k)$ that was constant over the distances within a category would be feasible with this imprecise distance information. This estimator will not always be PSD, unfortunately, since the spectral window corresponding to the step function space domain window (its Fourier transform) will be negative in some regions.

in the estimator b_τ for β yielding the feasible estimator of C :

$$\begin{aligned}\hat{C}_\tau = & \frac{1}{T_\tau} \sum_{j=0}^{L_M} \sum_{k=0}^{L_N} \sum_{m=j+1}^M \sum_{n=k+1}^N K_{MN}(j, k) \left(Y_{m,n}(b_\tau) Y_{m-j, n-k}(b_\tau)' + \right. \\ & \left. - \frac{1}{T_\tau} \sum_{m=1}^M \sum_{n=1}^N Y_{m,n}(b_\tau) Y_{m,n}(b_\tau)' \right).\end{aligned}\quad (3.13)$$

A set of sufficient conditions for consistent estimation of C includes restrictions on the weighting array in C1; a strengthening of moment and mixing assumptions in C2; and, in C3, a set of conditions on g and its derivatives to ensure consistency despite using b_τ in place of β .

C1. The $K_{MN}(j, k)$ are uniformly bounded, and $K_{MN}(j, k) \rightarrow 1$ as $\tau \rightarrow \infty$ ($M, N \rightarrow \infty$); $L_N = o(N^{1/3})$ and $L_M = o(M^{1/3})$.

C2. For some $\delta > 0$, $E(\|g(X_s; \beta)\|)^{4+\delta} < \infty$ and for both the X_s and W_s processes the mixing coefficient $\alpha_{\infty, \infty}(m)^{\delta/(2+\delta)} = o(m^{-4})$.

C3. $E \sup_B \|Y_{m,n}(b)\|^2 < \infty$ and $E \sup_B \|\partial/\partial b [Y_{m,n}(b)]\|^2 < \infty$.

The following proposition establishes consistency of $\hat{C}_\tau$.

Proposition 3. Given conditions A1–A3, B1–B5, and C1–C3, $\hat{C}_\tau \rightarrow C$ in probability as $\tau \rightarrow \infty$.

Proof. See the appendix.

3.3.1. PSD covariance matrix estimates

The weights $K_{MN}(j, k)$ can be chosen so as to guarantee PSD point estimates. One way to construct such weights involves using a spectral representation of the stationary process $Y_{m,n}(\beta)$ that is analogous to those for time series. V is proportional to the spectral density matrix at frequency zero of the process $Y_{m,n}(\beta)$ and the estimator $\hat{V}$ can be expressed as a weighted periodogram spectral density estimator. The periodogram is PSD, so if the weights $K_{MN}(j, k)$ correspond to a non-negative spectral window then the estimate will be PSD.

Weights satisfying these conditions can be easily made by the product of usual time domain weights. Consider for example the two-dimensional weight function that is a Bartlett window (Newey–West) in each dimension:

$$K_{MN}(j, k) = \begin{cases} (1 - |j|/L_M)(1 - |k|/L_N) & \text{for } |j| < L_M, |k| < L_N, \\ 0 & \text{else.} \end{cases}\quad (3.14)$$

Its Fourier transform is non-negative, so with these weights $\hat{C}_\tau$ will be PSD.

4. Estimation when economic distances are measured with error

In this section, locations s_i are allowed to be real-valued locations in $\mathbb{R}^2$ instead of being confined to an integer lattice. However, it is assumed that there is a minimum economic distance, d_0 , between agents, which was implicit when locations were confined to an integer lattice. While this assumption is made to simplify the presentation of mixing conditions, it can have a nice economic interpretation. If economic distances are identified with transactions costs between individuals, it has an economic justification as resulting from a fixed component of transaction costs.

Allowing real-valued economic distances facilitates the modelling of imperfectly measured distances. As mentioned above, imperfect measures of economic distance are likely to be the rule rather than the exception. We wish to allow this model to nest the case where errors in agents' locations are independent and measured economic distances are non-zero for distinct observations. Displacements on an integer lattice could not be independent without the possibility of there being more than one agent measured per lattice location.

Even with known distances, there are substantial gains from using the continuous space model in this section if, in addition to β , the covariance structure of the underlying random fields is of interest.¹² For example, it will enable non-parametric estimation of covariance functions using local averages. Typical samples will have few, if any, interpoint distances that are equal. Therefore, unlike a discrete time series where there are $T-k$ k th-order autocovariance observations, sample averages of k th-order spatial autocovariances of a lattice process will have very little data. Thus, non-parametric estimation of covariances of a discrete-index random field using a local average is infeasible. When the data is modelled as a continuous-index random field, its autocovariance function can be estimated as a local average if it is sufficiently smooth (see Hall et al., 1992, 1994).

Measurement error in distances will have no effect upon distribution theory below, besides the change in data model, as all the regularity conditions based on the true real-valued distances will still hold. It will however, have an impact upon covariance matrix estimators.

4.1. Data generation

The econometrician's sample still consists of locations $\{s_i\}$ within the sample region A_n , but s_i are now allowed to be real-valued vectors. The values of attributes of individual i are modelled as being determined by the value of an

¹² Continuous-index random field models are routinely used in geostatistics, see, e.g., Cressie (1991) for an overview.

underlying continuous-index random field $X(s)$ at location s_i . Again following Clark (1973), the observable random field on the irregular lattice H , X_{s_i} , is said to be subordinated to the continuous-index random field $X(s)$ and the process generating H is called the directing process. In addition to observations on X_{s_i} , the econometrician has observations on agents' locations that may be measured with error in a manner detailed below.

4.1.1. Continuous-index random field

The value of random variables at each location s_i is determined by a $\ell \times 1$ continuous-index random field $X(s)$ with the index $s \in \mathbb{R}^2$. The $X(s)$ are assumed to be stationary, i.e. the joint distribution of $X(s)$ for any collection of indices s is invariant to a shift in the indices. The process $X(s)$ is also assumed to be mixing.

4.1.2. Directing process

The irregular lattice H is modelled as being a realization of a point process on $\mathbb{R}^2$. A point process can be formally defined as a random counting measure.¹³ Let $(\Omega, \mathcal{G}, \mu)$ be a probability space and $\mathcal{P}$ be the collection of all locally finite counting measures on $\mathbb{R}^2$. Let $\mathcal{C}$ be the smallest sigma algebra generated by sets of the form $\{\phi \in \mathcal{P}: \phi(A) = n\}$ for all sets A in the Borel sigma algebra on $\mathbb{R}^2$ and all $\{n = 0, 1, 2, \dots\}$. A point process Φ on $\mathbb{R}^2$ is a measurable mapping from $(\Omega, \mathcal{G})$ into $(\mathcal{P}, \mathcal{C})$. Thus, a realization of a point process is counting measure $\phi(A)$ and H is the set of all the points in $\mathbb{R}^2$ whose locations it describes.

Six assumptions about this point process are maintained throughout. (D1), $X(s)$ and the point process Φ are assumed to be independent. (D2), Φ is simple which means that w.p.1 there are either zero points or one point at each location in $\mathbb{R}^2$. (D3), $0 < E\Phi(A) < \infty$ all A with finite Lebesgue measure. (D4), the assumption of a minimum economic distance requires Φ be assumed to be a 'hard core' point process meaning that w.p.1 there are no points within d_0 of each point in H . (D5), Φ is stationary which means that, letting $A + h$ denote the translation of A by $h \in \mathbb{R}^2$, the joint distribution of $\Phi(A_1 + h), \Phi(A_2 + h), \dots, \Phi(A_m + h)$ does not depend on h . (D6) Φ is assumed to be mixing. For simplicity we assume its mixing rate can be related to those for our random fields as follows. For a set of squares A_p covering $\mathbb{R}^2$ indexed by their corner coordinates p , define a random field $\psi_p \equiv \Phi(A_p)$. Φ is assumed mixing at a rate such that ψ_p satisfies C2.¹⁴

The restrictions on Φ do not exclude processes for which the probability of points in a region is increasing in the number of points nearby, cluster processes.

¹³ See e.g. Daley and Vere-Jones (1988) for a more complete presentation of this and alternative characterizations of point processes.

¹⁴ The process Φ need not be required to be mixing for the central limit results. The subordinated process will still satisfy the necessary covariance inequalities for the application of Bolthausen's central limit theorem when Φ is only ergodic.

Cluster processes that satisfy the minimum distance assumption are allowed. An independent sample of agents who resided in clusters, say in cities, would correspond to such a point process Φ .

4.1.3. Measurement error model

We model errors in economic distances as erroneously measured positions $\{s_i\}$. The baseline model of measurement error is that true locations $\{s_i\}$ are distorted by bounded position measurement errors. In other words, distance measurements are as though they come from the configuration $\{s_i + \varepsilon(s_i)\}$ where $\varepsilon(s_i)$ is a position location error.¹⁵ We model $\varepsilon(s_i)$ as being determined an underlying continuous index random field $\varepsilon(s)$ so that a given location s_i is associated with variables $X(s_i)$ and position measurement error $\varepsilon(s_i)$. The $\varepsilon(s_i)$ errors are not necessarily independent from X nor the point process Φ determining sampled locations. However we require them to be mixing and bounded, and that the conditional distribution of $X(s)$ given $\varepsilon(s)$ is well behaved, so that the following two conditions are satisfied:

E1. Measurement errors are bounded and the processes $X(s)$, Φ , and $\varepsilon(s)$ are jointly mixing, with mixing coefficients that satisfy condition C2.

E2. Letting $\mathfrak{T}_\varepsilon$ be the σ -algebra generated by the entire $\varepsilon(s)$ process, $E\{g(X(s); \beta) | \mathfrak{T}_\varepsilon\} < \infty$ for all s .

These conditions do not rule out measured distances for distinct observations being zero. For example, this situation could arise if economic distances between individuals corresponded to travel times between residences that were only measured up to city of residence. In such circumstances, measured distances between individuals who lived in the same city would not be available and might be taken to be zero. The covariance matrix estimator we present below will be consistent in this case, however, our technique for constructing PSD estimators will not work with measured distances that do not imply distinct locations. The covariance matrix estimator we present can easily be made to be PSD if observations' measured locations are distinct. Thus, it may be sensible to arbitrarily assign nearby but distinct locations to observations within a city in this example.

4.2. Estimation problem

The estimation problem remains the same as above with lattice locations, The GMM estimator of b_* is the argmin of Eq. (3.3) over the compact parameter

¹⁵ If measured economic distances do not correspond to a configuration in a low-dimensional Euclidean space, then a method called multidimensional scaling (see Mardia et al., 1979) can be used to fit a Euclidean configuration whose distances approximate the non-Euclidean distances.

space $\mathcal{B}$. The approach used above for regular lattice processes can be utilized to derive the large sample properties of b_τ with real-valued locations s_i . The trick is to define a regular lattice process whose partial sums approximate $\sum_{i=1}^{T_i} g(X_{s_i}; b)$.

4.2.1. Construction of regular lattice random field

The subordinated process X_{s_i} is used to define three random fields with coordinates from a square regular lattice.¹⁶ This regular lattice is constructed by dividing up $\mathfrak{R}^2$ into $d^* \times d^*$ squares whose diagonal is less than d_0 . Since points in H are at least d_0 from each other, there will be at most one point of H within each of these squares. Each square can then be assigned two integer coordinates contained in a vector p . Without loss of generality, we normalize the distances d^* to equal one in this section and the label p of each square is taken as the coordinates of its southwest corner.

Three new random fields can be defined on points of this regular lattice. The point process Φ defines the first process W_p as follows:

$$W_p \equiv \begin{cases} 1 & \text{if there is an } s_i \text{ in square } p, \\ 0 & \text{if not.} \end{cases}$$

The second process involves specifying a ‘latent’ process $\tilde{X}_p$:

$$\tilde{X}_p \equiv \begin{cases} X_{s_i} & \text{if there is an } s_i \text{ in square } p, \\ X(p) & \text{if not.} \end{cases}$$

In other words, when there is no s_i in square p the value of the underlying continuous index random field at the corner point of the square p is taken as the value of the process. Finally define a process $Y_p(b)$ as

$$Y_p(b) \equiv \begin{cases} (X_{s_i}; b) & \text{if there is an } s_i \text{ in square } p, \\ 0 & \text{if not.} \end{cases}$$

Fig. 2 illustrates this division of $\mathfrak{R}^2$ used to define these processes.

The new process Y_p is observed at coordinates p for all the squares completely within A_τ . Let A_τ^* be this observed region comprised of squares inside of A_τ . Let $|A_\tau^*|$ be the number of squares in A_τ^* and let $E\Phi$ on a square be denoted by λ .

¹⁶ This approach borrows heavily from the strategy in Guyon (1987) of relabelling a process on an irregular lattice with coordinates from a regular lattice.

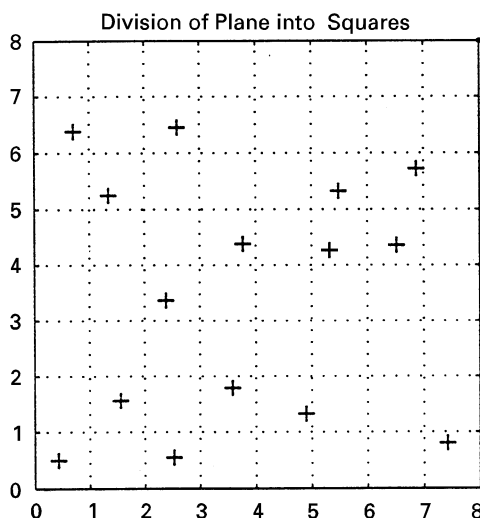


Fig. 2. Relabelling observations by e.g. the SW coordinate of their square maps observations on H to a regular lattice.

Defining these random fields is useful because sums of realizations of the random field $Y_p(b)$ correspond to sums of $g(X_{s_i}; b)$ plus a negligible number of boundary points:

$$\sum_{i=1}^{T_\tau} g(X_{s_i}; b) = \sum_{p \in A_\tau^*} Y_p(b) + \sum_{s_i \in A_\tau - A_\tau^*} g(X_{s_i}; b). \quad (4.1)$$

The strategy to get a law of large numbers for $\sum_{i=1}^{T_\tau} g(X_{s_i}; b)$ and a central limit result for $\sum_{i=1}^{T_\tau} g(X_{s_i}; \beta)$ is to get these results for $\sum_{p \in A_\tau^*} Y_p(b)$ and $\sum_{p \in A_\tau^*} Y_p(\beta)$ and demonstrate that the normalized partial sum of boundary points, $\sum_{s_i \in A_\tau - A_\tau^*} g(X_{s_i}; b)$, converges in probability to zero.

4.2.2. Consistency and asymptotic normality of b_τ

The regularity conditions are essentially the same as those above for the case where locations were constrained to be on an integer lattice. These results are stated as a separate proposition because the proofs are slightly different due to the need to keep track of boundary points when sums of observations in A_τ are approximated by sums of observations in A_τ^* , the constructed regular lattice.

Proposition 4. Given that conditions A1–A4 and B1–B5 hold for $X(s)$ with $V = \sum_{p \in \mathbb{Z}^2} \text{cov}(Y_0(\beta), Y_p(\beta))$, and D1–D6, then, as $\tau \rightarrow \infty$, $b_\tau \rightarrow \beta$ in probability and $\sqrt{T_\tau}(b_\tau - \beta) \Rightarrow N(0, D_0 \lambda^{-1} V D_0')$, where D_0 is as defined above in Eq. (3.11).

Proof. See the appendix.

4.3. Covariance matrix estimation

Covariance matrix estimators are affected by measurement errors in economic distances. If the minimum economic distance, d_0 , were known and distances were not measured with error, then the strategy above for relabelling would be feasible. The plane could be divided into $d^* \times d^*$ squares for a sufficiently small d^* . These squares could be given integer labels, a weighting array $K_{MN}(j, k)$ chosen, and the consistent estimator $\hat{C}_\tau$ constructed. This strategy is also viable when d_0 is approximated and there is bounded measurement error in distances.

A consistent estimator can be constructed using the following approach. First, choose an arbitrary small distance $\tilde{d}$ and divide the sample region into squares with sides of length $\tilde{d}$. Then label the squares with integer coordinates, referring to the coordinates of each square as $(\tilde{m}, \tilde{n})$. For simplicity we consider weighting arrays that can be written as a ‘scale parameter’ window:

$$K_{MN}(j, k) = \begin{cases} K(j/L_M, k/L_N) & \text{for } |j| < L_M, |k| < L_N, \\ 0 & \text{otherwise,} \end{cases}$$

where the function $K(\cdot, \cdot)$ is a bounded, continuous function on the rectangle $[-1, 1] \times [-1, 1]$ with absolutely summable Fourier coefficients, and $K(0, 0) = 1$. Weighting arrays that satisfy this restriction include the Bartlett window in Eq. (3.14) and many others (see, e.g., Priestley (1981) for other examples). We also take the sample region to be rectangular as above for simplicity.

If the distance $\tilde{d}$ can be chosen small enough so that there are not multiple observations in the same square, the resulting estimator can be written as:

$$\begin{aligned} \tilde{C}_\tau = & \frac{1}{T_\tau} \sum_{j=0}^{L_M} \sum_{k=0}^{L_N} \sum_{\tilde{m}=j+1}^{\tilde{M}} \sum_{\tilde{n}=k+1}^{\tilde{N}} K\left(\frac{j}{L_M}, \frac{k}{L_N}\right) \left[Y_{\tilde{m}, \tilde{n}}(b_\tau) Y_{\tilde{m}-j, \tilde{n}-k}(b_\tau)' \right. \\ & \left. + Y_{\tilde{m}-j, \tilde{n}-k}(b_\tau) Y_{\tilde{m}, \tilde{n}}(b_\tau)' \right] \\ & - \frac{1}{T_\tau} \sum_{\tilde{m}=j+1}^{\tilde{M}} \sum_{\tilde{n}=k+1}^{\tilde{N}} Y_{\tilde{m}, \tilde{n}}(b_\tau) Y_{\tilde{m}, \tilde{n}}(b_\tau)'. \end{aligned} \quad (4.2)$$

In the event that there are multiple observations with the same index (due to choice of large $\tilde{d}$ or non-distinct distance measurements), we take $\tilde{C}_\tau$ as given by an altered version of Eq. (4.2) with the bracketed terms containing all the cross

products for observations at a given distance. For example, if there are two points labelled $(\tilde{m}, \tilde{n})$, $Y_{\tilde{m}, \tilde{n}}(b_\tau)_1, Y_{\tilde{m}, \tilde{n}}(b_\tau)_2$ then the bracketed term will be

$$\left[Y_{\tilde{m}, \tilde{n}}(b_\tau)_1 Y_{\tilde{m}-j, \tilde{n}-k}(b_\tau)' + Y_{\tilde{m}-j, \tilde{n}-k}(b_\tau) Y_{\tilde{m}, \tilde{n}}(b_\tau)_1' + \right. \\ \left. Y_{\tilde{m}, \tilde{n}}(b_\tau)_2 Y_{\tilde{m}-j, \tilde{n}-k}(b_\tau)' + Y_{\tilde{m}-j, \tilde{n}-k}(b_\tau) Y_{\tilde{m}, \tilde{n}}(b_\tau)_2' \right].$$

The following proposition establishes consistency of $\tilde{C}_\tau$ for an arbitrary choice of $\tilde{d}$.

Proposition 5. *Given that conditions A1–A4, B1–B5, and C1–C3 hold for $X(s)$, conditions D1–D6, E1 and E2, then $\tilde{C}_\tau \rightarrow C$, in probability.*

Proof. See the appendix.

If each observation is measured as having a distinct location, we can easily construct a $\tilde{C}_\tau$ that will be PSD for a given sample. In this case, $\tilde{d}$ can be chosen small enough so that there is only one observation per square by, e.g., expressing coordinates with enough decimal places to ensure that no two observations have the same location. $\tilde{C}_\tau$ will then be positive definite for a given sample if the kernel $K(\cdot, \cdot)$ has a non-negative Fourier transform. If measurements of economic distances do not locate agents in distinct locations, this strategy to get PSD estimates cannot always be implemented. However, $\tilde{C}_\tau$ will still remain consistent as long as the measurement errors satisfy conditions E1 and E2.

Note that for a fixed choice of $\tilde{d}$, the sequence of estimators $\tilde{C}_\tau$ is not guaranteed to contain only positive-definite elements. This choice of $\tilde{d}$ (with an appropriate $K(\cdot, \cdot)$) simply constrains the first element of the sequence to be positive-definite. In principle, $\tilde{d}$ could be explicitly allowed to vary with τ to assure that $\tilde{C}_\tau$ was a sequence of positive-definite matrices. We retain the assumption of fixed $\tilde{d}$ because allowing it to vary with τ seems to have little practical value and a considerable cost in complicating the proof of Proposition 5, requiring additional assumptions about the measurement error process.

5. Empirical example

This section provides an illustration that our model of cross-sectional dependence is empirically relevant. In the example presented below, parameters are estimated by OLS, a just-identified GMM estimator. Therefore, the impact of allowing for dependence is only upon inference, not point estimates. We focus on standard errors for OLS point estimates to simplify comparisons between our estimates and those that do not allow for spatial correlation. However, the impact of allowing for dependence will likely be even greater in over-identified systems where estimated asymptotic covariance matrices will

determine weighting matrices and hence point estimates and tests of overidentifying restrictions as well as standard errors. Thus, by focusing on inference in this just-identified case, the examples below offer a relatively conservative illustration of the impact of allowing for spatial dependence.

The application is a growth regression from Conley and Ligon (1995). The measure of economic distance is constructed and likely to be measured with error. The measure of economic distance in this example is a measure of the costs of transporting physical capital between countries.

This growth regression example is an OLS regression with a specification and variables used by Barro (1991). The growth rate of real GDP from 1960 to 1985 is regressed on: initial values of real GDP (GDP60), secondary and primary school enrollment rates (SEC60 and PRIM60), the share of real government 'consumption' expenditures to real GDP for 1970–1985 (g^c/Y), revolutions from 1960 to 1985 (REV), assassinations per million (ASSASS), absolute deviation of 1960 PPP investment deflator from its sample mean (PPI60DEV), and region dummy variables for sub Saharan Africa and Latin America. The point of this regression is to examine whether growth rates are related to measures of human capital, political stability, fiscal variables, etc.

In Table 2 we present OLS point estimates and standard errors calculated in three different ways. First, usual OLS standard errors are in the column labeled IID SE. Heteroskedasticity consistent standard errors are in the column labelled HET SE (Eicker, 1967; Huber, 1967; White, 1980). Finally, standard errors calculated using the estimator $\hat{C}_\tau$ defined above that allow for spatial dependence as well as heteroskedasticity are in the column labelled Spatial SE. The regression reported in Table 2 used a cross section of 95 countries for whom data was available.

The measurement of economic distance used to construct these spatial standard errors is a measure of transportation cost of physical capital between countries. The construction of this metric is detailed in Conley and Ligon (1995) but to give an idea of relative transportation costs, Table 1 provides measures of this cost measurement between a few countries. The spatial standard errors in Table 1 were calculated using a window whose weights were equal to one for

Table 1

Example country pair	Economic distance measure
USA to Mexico	32
USA to Algeria	141
USA to Japan	119
UK to France	53
USA to Pakistan	218

Table 2

The dependent variable is growth rate of real GDP 1960–85

	Point estimate	IID S.E.	HET S.E.	Spatial S.E.
Constant	0.0333	0.0063	0.0070	0.0053
GDP60	– 0.0066	0.0010	0.0009	0.0008
SEC60	0.0124	0.0106	0.0077	0.0019**
PRIM60	0.0274	0.0060	0.0060	0.0065
g^c/Y	– 0.0959	0.0260	0.0269	0.0359
REV	– 0.0208	0.0085	0.0081	0.0072
ASSASS	– 0.0024	0.0029	0.0018	0.0009**
PPI60DEV	– 0.0139	0.0051	0.0048	0.0064*
Africa	– 0.0107	0.0038	0.0041	0.0046
Latin America	– 0.0137	0.0033	0.0032	0.0028

observations less than 75 units apart and zero for more distant observations. Some idea of the relative magnitude of this truncation value of 75 is afforded by Table 1. The qualitative results discussed below are robust to changes in this cutoff value and/or window specification. The same qualitative results were obtained using the truncated window with cutoffs from 60 to 90 as well as using the two-dimensional Bartlett window in equation (3.14) with truncation points between 75 and 225.¹⁷

The estimates in Table 2 provide evidence that allowing for spatial dependence can be important for conducting inference. Overall, the magnitude of the difference between spatial standard errors is at least as great as the difference between IID standard errors and HET standard errors. In particular, the magnitudes of the spatial standard errors are enough to change the values of t -tests for significance for the two double starred variables, SEC60 and ASSASS, and to impact the confidence interval enough to change the economic significance of the estimated coefficient on the single starred variable: PPI60DEV.

Table 2 illustrates another important point: spatial dependence does not imply that standard errors will rise. Indeed, six of nine spatial standard error estimates are smaller than their IID or HET counterparts. It is important to remember that asymptotic variances may be smaller with spatially dependent data, just as asymptotic variances can be lower for dependent time series averages than independent series averages.

¹⁷ Specifically, the results discussed below that are robust are: significance of the starred variables in Table 2; relative magnitudes of the differences between IID, HET, and spatial standard errors; and the fact that many of the spatial standard errors are smaller than IID or HET standard errors.

6. Conclusion

This paper presents practical estimation methods for econometric models of cross sectional dependence. The foremost feature of these models is that they allow the econometrician to use information on agents' interdependence – their economic distance – to characterize dependence structures without parametric assumptions. As described in the introduction, there are many applications where the econometrician may have such auxiliary economic distance information. The paper has presented consistency and distribution results for GMM estimators and consistency results for a class of PSD covariance matrix estimators that remain consistent in the presence of measurement errors in economic distances. These results enable estimation, inference, efficient estimation, and tests of overidentifying restrictions with dependent cross-sectional data.

The fundamental restriction placed on the structure of dependence in this paper is that it can be characterized by a configuration of points in a Euclidean space, presumably of a low dimension relative to the sample size. In many applications measures of transactions costs between agents are potential economic distances measures.¹⁸ When fixed costs are a large portion of agents' transaction costs then their variables' dependence structure will not be well represented in a low-dimensional Euclidean space. However, estimators of the form of those in this paper will still be feasible. Thus, obtaining large sample results for GMM and covariance matrix estimators when dependence is characterized by more general metrics may prove to be a quite useful extension.

The paper also imposes two important restrictions on the process determining which agents and locations are observed. The key restriction upon this directing process is that it is independent from the underlying random field, X . There are clearly applications that will violate this assumption. Allowing for endogenous locations is likely to require an explicit model of how locations and variables in the moment conditions are jointly determined.¹⁹ We also maintain the assumption that the sampling process has a minimum distance. This assumption is not necessary, but provides a simple way to insure that the stationary distribution is consistently estimated and simplifies the derivation of limiting distributions. Replacing this assumption with a more complicated one that allows for distances to be arbitrarily small while preserving consistency may prove a useful extension.²⁰

¹⁸ As in, for example, Conley and Tsiang (1994) where time costs of travel between agents' locations are used as economic distances.

¹⁹ Carrasco et al. (1998) present a such a model for subordinated time series.

²⁰ In particular, the sampling scheme in Hall and Patil (1994) might be used.

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Appendix A. Proofs of propositions

Proof of Proposition 1. A pointwise law of large numbers (LLN) for

$$\frac{1}{T_\tau} \sum_{i=1}^{T_\tau} g(X_{s_i}; b)$$

is available from the ergodic theorem for random fields. Using the process $Y_s(b)$ defined in Eq. (3.9),

$$\frac{1}{T_\tau} \sum_{i=1}^{T_\tau} g(X_{s_i}; b) = \frac{1}{T_\tau} \sum_{s \in A_\tau} Y_s(b) = \left[\frac{|A_\tau|}{T_\tau} \right] \left[\frac{1}{|A_\tau|} \sum_{s \in A_\tau} Y_s(b) \right]. \quad (\text{A.1})$$

An ergodic theorem can be applied to each of the bracketed terms in Eq. (A.1) to obtain a pointwise law of large numbers.

First consider $(1/|A_\tau|) \sum_{s \in A_\tau} Y_s(b)$. $Y_s(b)$ can be written as the product of stationary, ergodic processes $g(X_s; b)$ and W_s , so it is stationary and ergodic. The ergodic theorem for random fields (see e.g. Adler, 1981; Daley and Vere-Jones, 1988) implies

$$\frac{1}{|A_\tau|} \sum_{s \in A_\tau} Y_s(b) \rightarrow EY_s(b), \text{ w.p.1. as } \tau \rightarrow \infty.$$

Remembering that X_s and W_s are independent:

$$EY_s(b) = EW_s E g(X_{s_i}; b).$$

Next consider the term $|A_\tau|/T_\tau$ in Eq. (A.1). Its inverse can be written as the sample average of W_s :

$$\frac{T_\tau}{|A_\tau|} = \frac{1}{|A_\tau|} \sum_{s \in A_\tau} W_s.$$

The ergodic theorem implies that

$$\frac{1}{|A_\tau|} \sum_{s \in A_\tau} W_s \rightarrow EW_s, \text{ w.p.1. as } \tau \rightarrow \infty$$

using Slutsky's theorem:

$$\frac{|A_\tau|}{T_\tau} \rightarrow (EW_s)^{-1} \text{ w.p.1. as } \tau \rightarrow \infty.$$

Therefore

$$\frac{1}{T_\tau} \sum_{i=1}^{T_\tau} g(X_{s_i}; b) \rightarrow Eg(X_{s_i}; b) \text{ w.p.1. as } \tau \rightarrow \infty. \quad (\text{A.2})$$

This pointwise LLN, the first moment continuity of $g(X_{s_i}; b)$, and the compactness of the parameter space B can be used to get a uniform law of large numbers. The argument is identical to that in Hansen (1991) for the time-series case.

Thus Eq. (A.2) holds uniformly in b . Since $\Omega_\tau \rightarrow \Omega$ in probability, Eq. (3.3) implies that $J_T(b) \rightarrow J(b) \equiv Eg(X_{s_i}; b)' \Omega Eg(X_{s_i}; b)$ in probability, uniformly in b . $Eg(X_{s_i}; b) = 0$ only for $b = \beta$ so $J(b)$ is minimized only at $b = \beta$. Therefore, by standard arguments (e.g. Amemiya, 1984, Theorem 4.1.1) $b_\tau \rightarrow \beta$ in probability.

Proof of Proposition 2. A uniform LLN constructed as in the proof of Proposition 1 can be applied along with Slutsky's Theorem to show that

$$\begin{aligned} \frac{1}{T_\tau} \sum_{i=1}^{T_\tau} Dg(X_{s_i}; b_\tau) &\rightarrow EDg(X_{s_i}; \beta) \text{ in probability and} \\ \left\{ \left[\frac{1}{T_\tau} \sum_{i=1}^{T_\tau} Dg(X_{s_i}; b_\tau) \right]' \Omega_\tau \left[\frac{1}{T_\tau} \sum_{i=1}^{T_\tau} Dg(X_{s_i}; b_\tau) \right] \right\}^{-1} &\left[\frac{1}{T_\tau} \sum_{i=1}^{T_\tau} Dg(X_{s_i}; b_\tau) \right]' \Omega_\tau \\ &\rightarrow \{ EDg(X_{s_i}; \beta)' \Omega EDg(X_{s_i}; \beta) \}^{-1} EDg(X_{s_i}; \beta)' \Omega \end{aligned}$$

in probability.

Repeating Eq. (3.10) here for convenience

$$\frac{1}{\sqrt{T_\tau}} \sum_{i=1}^{T_\tau} g(X_{s_i}; \beta) = \sqrt{\frac{|A_\tau|}{T_\tau}} \frac{1}{\sqrt{|A_\tau|}} \sum_{s \in A_\tau} Y_s(\beta).$$

In the proof of Proposition 1 it is shown that $T_\tau/|A_\tau| \rightarrow EW_s = \lambda$ w.p.1. Therefore, using Slutsky's theorem, the factor $\sqrt{|A_\tau|/T_\tau} \rightarrow \lambda^{-1/2}$ w.p.1. The conclusion of the proposition follows upon using Bolthausen's (1982) central limit theorem to obtain the asymptotic distribution of $(1/\sqrt{|A_\tau|}) \sum_{s \in A_\tau} Y_s(\beta)$.

Mixing conditions in Bolthausen's (1982) central limit theorem use a metric of maximum coordinatewise distance, therefore the mixing coefficient defined above needs to be related to a mixing coefficient defined using this metric. Let $\mathfrak{A}$ be the σ -algebra generated by $X_{s_m}, s_m \in A$ with $A \subset H$, and let $|A|$ be the number of $s_m \in A$. Let $\pi(A_1, A_2)$ denote the maximum coordinatewise distance from an element of A_1 to an element of A_2 . Define a mixing coefficient using the metric π as

$$\begin{aligned} \tilde{\alpha}_{k,l}(n) &\equiv \sup\{|P(A \cap B) - P(A)P(B)|\}, A \in \mathfrak{A}_{A_1}, B \in \mathfrak{A}_{A_2} \text{ and} \\ |A_1| &\leq k, |A_2| \leq l, \pi(A_1, A_2) \geq n \end{aligned} \quad (\text{A.3})$$

Because maximum coordinatewise distance is weakly less than Euclidean distance and $\alpha_{k,l}(n)$ is decreasing in n :

$$\tilde{\alpha}_{k,l}(n) = \alpha_{k,l}(n)$$

Therefore conditions analogous to B1–B3 hold for $\tilde{\alpha}_{k,l}(n)$:

$$\text{B1'}. \sum_{m=1}^{\infty} m \tilde{\alpha}_{k,l}(m) < \infty \text{ for } k + l \leq 4.$$

$$\text{B2'}. \tilde{\alpha}_{1,\infty}(m) = o(m^{-2}).$$

$$\text{B3'}. \text{For some } \delta > 0, E(\|g(X_s; \beta)\|)^{2+\delta} < \infty \text{ and } \sum_{m=1}^{\infty} m \tilde{\alpha}_{1,1}(m)^{\delta/(2+\delta)} < \infty.$$

$Y_s(\beta)$ is the product of $g(X_{s_i}; \beta)$ and W_s and so is a stationary random field satisfying mixing conditions B1', B2', and B3' which are those required to apply Bolthausen's (1982) central limit theorem. The remaining conditions in Bolthausen (1982) are the non-singularity of V (B5) and a boundary condition satisfied by condition A1. Therefore, Bolthausen (1982) can be applied to an arbitrary linear combination of $Y_s(\beta)$ to imply the desired result via the Cramer–Wold device.

Proof of Proposition 3. First we show that

$$\hat{V} \rightarrow \sum_{k=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} E Y_{0,0}(\beta) Y_{k,j}(\beta)' \text{ in prob. as } \tau \rightarrow \infty. \quad (\text{A.4})$$

The proof is a generalization of the time-series analog in Quah (1990). Since any sequence of symmetric matrices A_τ converges to a symmetric matrix A if and only if $\kappa' A_\tau \kappa \rightarrow \kappa' A \kappa$ for any vector κ , the proposition for vector $Y_{m,n}$ will follow from considering an arbitrary linear combination of $Y_{m,n}$. Therefore we restrict attention to the case of scalar $Y_{m,n}(\beta)$. Thus, we first show

$$\begin{aligned} & \frac{2}{MN} \sum_{j=0}^{L_M} \sum_{k=0}^{L_N} \sum_{m=j+1}^M \sum_{n=k+1}^N K_{MN}(j, k) Y_{m,n}(\beta) Y_{m-j, n-k}(\beta) \\ & - \frac{1}{MN} \sum_{m=1}^M \sum_{n=1}^N [Y_{m,n}(\beta)]^2 \\ & \rightarrow \sum_{k=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} E Y_{0,0}(\beta) Y_{k,j}(\beta) \text{ in probability.} \end{aligned}$$

To simplify notation let $Y_{m,n} = 0$ for non-positive values of either index. Define an array $Z_{MN,mn}$:

$$Z_{MN,mn} = \sum_{j=0}^{L_M} \sum_{k=0}^{L_N} K_{MN}(j, k) [Y_{m,n}(\beta) Y_{m-j, n-k}(\beta) - E Y_{m,n}(\beta) Y_{m-j, n-k}(\beta)].$$

The first step is to prove a lemma analogous to Quah (1990, Theorem 3.3).

Lemma A.

$$\frac{1}{MN} \sum_{m=1}^M \sum_{n=1}^N Z_{MN,mn} \rightarrow 0 \text{ in probability.}$$

Proof. The strategy is to get a bound on $\text{var}\{\sum_{m=1}^M \sum_{n=1}^N Z_{MN,mn}\}$ and then use Chebyshev's inequality to get a convergence in probability result:

$$\text{var}\left\{\sum_{m=1}^M \sum_{n=1}^N Z_{MN,mn}\right\} = \left| \sum_{m=1}^M \sum_{n=1}^N \sum_{m'=1}^M \sum_{n'=1}^N E Z_{MN,mn} Z_{MN,m'n'} \right|$$

The triangle inequality implies

$$\begin{aligned} \text{var} \left\{ \sum_{m=1}^M \sum_{n=1}^N Z_{MN,mn} \right\} &\leq \sum_{m=1}^M \sum_{n=1}^N \sum_{|m-m'| \leq 2L_M} \sum_{|n-n'| \leq 2L_N} |EZ_{MN,mn} Z_{MN,m'n'}| \\ &+ \sum_{m=1}^M \sum_{n=1}^N \sum_{m',n': |m-m'| > 2L_M \text{ or } |n-n'| > 2L_N} |EZ_{MN,mn} Z_{MN,m'n'}| \end{aligned} \quad (\text{A.5})$$

having divided the terms into close ones (within $2L_M$ and $2L_N$ in each direction) and far ones (farther than $2L_M$ or $2L_N$). Note that if the sample region were not rectangular, the $EZ_{MN,mn} Z_{MN,m'n'}$ terms could still be divided into close and far groups of terms.

First look at the close terms. No matter what the shape of the sample region, the maximum number of points within $2L_M, 2L_N$ of a point is $(4L_M + 1)(4L_N + 1)$. Therefore,

$$\begin{aligned} &\sum_{m=1}^M \sum_{n=1}^N \sum_{|m-m'| < 2L_M} \sum_{|n-n'| < 2L_N} |EZ_{MN,mn} Z_{MN,m'n'}| \\ &\leq MN(4L_M + 1)(4L_N + 1) \sup_{1 \leq m' \leq M, 1 \leq n' \leq N} \|Z_{MN,m'n'}\|_2^2 \end{aligned}$$

The next step is to bound $\sup_{1 \leq m' \leq M, 1 \leq n' \leq N} \|Z_{MN,m'n'}\|_2^2$. Minkowski's inequality implies:

$$\begin{aligned} \|Z_{MN,mn}\|_q &\leq \sum_{j=0}^{L_M} \sum_{k=0}^{L_N} K_{MN}(j, k) \|Y_{m,n}(\beta) Y_{m-j, n-k}(\beta)\|_q \\ &- EY_{m,n}(\beta) Y_{m-j, n-k}(\beta) \|_q \end{aligned} \quad (\text{A.6})$$

Assumption C2 and $W_{m,n}$ being a dummy variable independent of X imply $Y_{m,n}(\beta)$ has finite $(4 + \delta)$ th moments which implies that

$$\sup_{j,k} \|Y_{m,n}(\beta) Y_{m-j, n-k}(\beta) - EY_{m,n}(\beta) Y_{m-j, n-k}(\beta)\|_q < \infty \quad \text{for } q = 2, 2 + \delta.$$

This, along with the assumption that $K_{MN}(j, k)$ are bounded uniformly, implies that

$$\|Z_{MN,mn}\|_q \leq c_1(L_M + 1)(L_N + 1)$$

for $q = 2, 2 + \delta$ and some constant c_1 , thus giving the following bounds:

$$\|Z_{MN,mn}\|_2^2 \leq c_1^2 (L_M + 1)^2 (L_N + 1)^2$$

and

$$\|Z_{MN,mn}\|_{2+\delta}^2 \leq c_1^2 (L_M + 1)^2 (L_N + 1)^2. \quad (\text{A.7})$$

Therefore the near terms satisfy

$$\begin{aligned} & \sum_{m=1}^M \sum_{n=1}^N \sum_{|m-m'| < 2L_M} \sum_{|n-n'| < 2L_N} |\text{E} Z_{MN,mn} Z_{MN,m'n'}| \\ & \leq c_1^2 MN (4L_M + 1) (4L_N + 1) (L_M + 1)^2 (L_N + 1)^2. \end{aligned} \quad (\text{A.8})$$

Next consider the far apart terms.

$$\sum_{m=1}^M \sum_{n=1}^N \sum_{\substack{m', n': |m-m'| > 2L_M \text{ or } |n-n'| > 2L_N}} |\text{E} Z_{MN,mn} Z_{MN,m'n'}|.$$

We use the following mixing inequality, it is analogous to Bolthausen (1982), Lemma 1) and also to Ibragimov and Linnik (1971, Theorems 17.2.2 and 17.2.3). The mixing coefficient $\tilde{\alpha}$ is as defined in the proof of Proposition 2.

Inequality A. If X_s is a mixing random field with finite $(2 + \delta)$ th moments satisfying B1'–B3', then

$$|\text{cov}(X_s, X_{s'})| \leq c \tilde{\alpha}_{\infty, \infty} (\pi(s, s'))^{\delta/(2+\delta)} \|X_s\|_{2+\delta} \|X_{s'}\|_{2+\delta}$$

for some constant c that may depend on δ .

Applying Inequality A yields

$$|\text{E} Z_{MN,mm} Z_{MN,m'n'}| \leq c_2 \tilde{\alpha}_{\infty, \infty} (\min(L_M, L_N))^{\delta/(2+\delta)} \sup_{m,n} \|Z_{MN,mn}\|_{2+\delta}^2$$

for some constant c_2 .

The greatest number of far away terms would result from a sample region very nearly a line, where there would be $MN - 4 \min(L_M, L_N) - 1$ terms at most.

Therefore the far terms can be bounded as follows:

$$\begin{aligned} & \sum_{m=1}^M \sum_{n=1}^N \sum_{m', n': |m-m'| > 2L_M \text{ or } |n-n'| > 2L_N} |EZ_{MN, mn} Z_{MN, m'n'}| \\ & < c_2 MN (MN - 4 \min(L_M, L_N) - 1) \tilde{\alpha}_{\infty, \infty} (\min(L_M, L_N))^{\delta/(2+\delta)} \\ & \sup \|Z_{MN, mn}\|_{2+\delta}^2 < c_2 (MN)^2 \tilde{\alpha}_{\infty, \infty} (\min(L_M, L_N))^{\delta/(2+\delta)} \sup \|Z_{MN, mn}\|_{2+\delta}^2. \end{aligned}$$

Using Eq. (A.7),

$$\begin{aligned} & \sum_{m=1}^M \sum_{n=1}^N \sum_{m', n': |m-m'| > 2L_M \text{ or } |n-n'| > 2L_N} |EZ_{MN, mn} Z_{MN, m'n'}| \\ & \leq c_3 (MN)^2 \tilde{\alpha}_{\infty, \infty} (\min(L_M, L_N))^{\delta/(2+\delta)} (L_M + 1)^2 (L_N + 1)^2 \end{aligned} \quad (\text{A.9})$$

for yet another constant c_3 .

Putting together the far and close terms from Eqs. (A.8) and (A.10):

$$\begin{aligned} \text{var} \left\{ \sum_{m=1}^M \sum_{n=1}^N Z_{MN, mn} \right\} & \leq c_1^2 \{ MN (4L_M + 1) (4L_N + 1) (L_M + 1)^2 (L_N + 1)^2 \\ & + c_3 (MN)^2 \tilde{\alpha}_{\infty, \infty} (\min(L_M, L_N))^{\delta/(2+\delta)} (L_M + 1)^2 (L_N + 1)^2 \}. \end{aligned} \quad (\text{A.10})$$

Now using Chebychev's inequality:

$$\begin{aligned} \Pr \left(\left| \frac{1}{MN} \sum_{m=1}^M \sum_{n=1}^N Z_{MN, mn} \right| \geq \varepsilon \right) & \leq \frac{1}{\varepsilon^2} \frac{1}{(MN)^2} \text{var} \left(\sum_{m=1}^M \sum_{n=1}^N Z_{MN, mn} \right) \\ & \leq \frac{1}{\varepsilon^2} \frac{1}{MN} c_1^2 (4L_M + 1) (4L_N + 1) (L_M + 1)^2 (L_N + 1)^2 \\ & + \frac{1}{\varepsilon^2} c_3 \tilde{\alpha}_{\infty, \infty} (\min(L_M, L_N))^{\delta/(2+\delta)} (L_M + 1)^2 (L_N + 1)^2 \end{aligned}$$

Since $L_M = o(M^{1/3})$, $L_N = o(N^{1/3})$ and $\tilde{\alpha}_{\infty, \infty}(m)^{\delta/(2+\delta)} = o(m^{-4})$ and A_τ grows uniformly in two non-opposing directions (M and N are of same order), both of the terms on the right-hand side go to zero and Lemma A follows.

Note that $\hat{V} - E\hat{V} = (2/MN) \sum_{m=1}^M \sum_{n=1}^N Z_{MN, mn} - \{(1/MN) \sum_{m=1}^M \sum_{n=1}^N Y_{m,n}^2(\beta) - EY_{m,n}^2(\beta)\}$. Lemma A applied to the first term and the LLN derived in the proof of Proposition 1 applied to the second term imply that $\hat{V} - E\hat{V} \rightarrow 0$ in probability.

The final step is to show that $E\hat{V}$ gets close to V :

$$\begin{aligned} E\hat{V} &\equiv \frac{2}{MN} \sum_{j=0}^{L_M} \sum_{k=0}^{L_N} \sum_{m=j+1}^M \sum_{n=k+1}^N K_{MN}(j, k) EY_{m,n}(\beta) Y_{m-j, n-k}(\beta) \\ &\quad - \frac{1}{MN} \sum_{m=1}^M \sum_{n=1}^N EY_{m,n}(\beta) Y_{m,n}(\beta) \end{aligned} \quad (\text{A.11})$$

rewriting

$$\begin{aligned} E\hat{V} - var\left\{\frac{1}{\sqrt{MN}} \sum_{m=1}^M \sum_{n=1}^N Y_{m,n}(\beta)\right\} &= \frac{2}{MN} \sum_{j=0}^{L_M} \sum_{k=0}^{L_N} (K_{MN}(j, k) - 1) \\ &\quad \sum_{m=j+1}^M \sum_{n=k+1}^N EY_{m,n}(\beta) Y_{m-j, n-k}(\beta) \\ &\quad - \frac{1}{MN} \sum_{j=L_M+1}^{M-1} \sum_{k=L_N+1}^{N-1} \sum_{m=j+1}^M \sum_{n=k+1}^N EY_{m,n}(\beta) Y_{m-j, n-k}(\beta). \end{aligned}$$

Using Inequality A to bound the first term:

$$\begin{aligned} &\left| \frac{2}{MN} \sum_{j=0}^{L_M} \sum_{k=0}^{L_N} (K_{MN}(j, k) - 1) \sum_{m=j+1}^M \sum_{n=k+1}^N EY_{m,n}(\beta) Y_{m-j, n-k}(\beta) \right| \\ &\leq c_4 \sum_{j=0}^{L_M} \sum_{k=0}^{L_N} \left| \frac{(M-j)(N-k)}{MN} (K_{MN}(j, k) - 1) \right| \tilde{\alpha}_{\infty, \infty}(\max(j, k))^{\delta/(2+\delta)} \\ &\leq c_4 \sum_{j=0}^{L_M} \sum_{k=0}^{L_N} |(K_{MN}(j, k) - 1)| \tilde{\alpha}_{\infty, \infty}(\max(j, k))^{\delta/(2+\delta)} \end{aligned} \quad (\text{A.12})$$

for some constant c_4 . The assumption that $\tilde{\alpha}_{\infty, \infty}(m)^{\delta/(2+\delta)} = o(m^{-4})$ implies $\sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \tilde{\alpha}_{1,1}(\max(j, k))^{\delta/(2+\delta)} < \infty$. Therefore, since $K_{MN}(j, k) \rightarrow 1$ for each j, k the dominated convergence theorem implies

$$c_4 \sum_{j=0}^{L_M} \sum_{k=0}^{L_N} |(K_{MN}(j, k) - 1)| \tilde{\alpha}_{\infty, \infty}(\max(j, k))^{\delta/(2+\delta)} \rightarrow 0.$$

Inequality A used on the second term implies that for some constant c_5 :

$$\begin{aligned} & \left| \frac{1}{MN} \sum_{j=L_M+1}^{M-1} \sum_{k=L_N+1}^{N-1} \sum_{m=j+1}^M \sum_{n=k+1}^N E Y_{m,n}(\beta) Y_{m-j,n-k}(\beta) \right| \\ & \leq c_5 \sum_{j=L_M+1}^{M-1} \sum_{k=L_N+1}^{N-1} \frac{(M-j)(N-k)}{MN} \tilde{\alpha}_{\infty, \infty} (\max(j, k))^{\delta/(2+\delta)} \\ & \leq c_5 \sum_{j=L_M+1}^{M-1} \sum_{k=L_N+1}^{N-1} \tilde{\alpha}_{\infty, \infty} (\max(j, k))^{\delta/(2+\delta)}. \end{aligned}$$

As the right-hand side is a constant times the tail of a convergent double sequence, it converges to zero. Therefore result Eq. (A.4) follows.

Now, we must account for the estimated parameters in $\hat{C}_\tau$. The strategy mirrors that in Andrews (1991). Define

$$\begin{aligned} \Gamma(b) &= \frac{2}{MN} \sum_{j=0}^{L_M} \sum_{k=0}^{L_N} \sum_{m=j+1}^M \sum_{n=k+1}^N K_{MN}(j, k) Y_{m,n}(b) Y_{m-j,n-k}(b) \\ &\quad - \frac{1}{MN} \sum_{m=1}^M \sum_{n=1}^N [Y_{m,n}(b)]^2, \\ \Gamma(\beta) &= \Gamma_1(b) + \Gamma_2(b). \end{aligned}$$

Thus, $\hat{V} = \Gamma(\beta)$ and the feasible estimator $\hat{C}_\tau = (T_\tau/MN)^{-1} \Gamma(b_\tau)$. In the proof of proposition 1 it was shown that $(T_\tau/MN)^{-1} \rightarrow \lambda$ in prob. and we have result (A.4); therefore, the proposition follows from a demonstration that $\Gamma(b_\tau) - \Gamma(\beta) \rightarrow 0$ in probability.

The first step is to take a mean-value expansion of $\Gamma_1(b_\tau)$ and $\Gamma_2(b_\tau)$ around β as is done in Andrews (1991):

$$\begin{aligned} \frac{\sqrt{MN}}{L_M L_N} (\Gamma_1(b_\tau) - \Gamma_1(\beta)) &= \frac{1}{L_M L_N} \frac{\partial}{\partial b} \Gamma_2(\bar{b}) \sqrt{MN} (b_\tau - \beta) \\ &= \sqrt{MN} (b_\tau - \beta) \frac{1}{L_M L_N} \sum_{j=0}^{L_M} \sum_{k=0}^{L_N} K_{MN}(j, k) \frac{2}{MN} \\ &\quad \times \sum_{m=j+1}^M \sum_{n=k+1}^N \frac{\partial}{\partial b} [Y_{m,n}(\bar{b}) Y_{m-j,n-k}(\bar{b})] \end{aligned}$$

for some $\bar{b}$ between b_τ and β .

The next step is to get a bound on $(1/MN)\sum_{m=j+1}^M\sum_{n=k+1}^N(\partial/\partial b)[Y_{m,n}(\bar{b})Y_{m-j,n-k}(\bar{b})]$. Writing out the derivative and applying the triangle inequality yields

$$\begin{aligned} & \left\| \frac{1}{MN} \sum_{m=j+1}^M \sum_{n=k+1}^N \left\{ \frac{\partial}{\partial b}[Y_{m,n}(\bar{b})]Y_{m-j,n-k}(\bar{b}) + Y_{m,n}(\bar{b}) \frac{\partial}{\partial b}[Y_{m-j,n-k}(\bar{b})] \right\} \right\| \\ & \leq \frac{1}{MN} \sum_{m=j+1}^M \sum_{n=k+1}^N \left\| \frac{\partial}{\partial b}[Y_{m,n}(\bar{b})]Y_{m-j,n-k}(\bar{b}) \right\| \\ & \quad + \frac{1}{MN} \sum_{m=j+1}^M \sum_{n=k+1}^N \left\| Y_{m,n}(\bar{b}) \frac{\partial}{\partial b}[Y_{m-j,n-k}(\bar{b})] \right\| \end{aligned}$$

Each term in these sums can be bounded by

$$\left\| \frac{\partial}{\partial b}[Y_{m,n}(\bar{b})]Y_{m-j,n-k}(\bar{b}) \right\| \leq \sup_B \|Y_{m-j,n-k}(b)\| \sup_B \left\| \frac{\partial}{\partial b}[Y_{m,n}(b)] \right\|.$$

Using this bound for each term:

$$\begin{aligned} & \left\| \frac{1}{MN} \sum_{m=j+1}^M \sum_{n=k+1}^N \frac{\partial}{\partial b}[Y_{m,n}(\bar{b})]Y_{m-j,n-k}(\bar{b}) \right\| \\ & \leq \frac{1}{MN} \sum_{m=j+1}^M \sum_{n=k+1}^N \sup_B \|Y_{m-j,n-k}(b)\| \sup_B \left\| \frac{\partial}{\partial b}[Y_{m,n}(b)] \right\| \\ & \quad + \frac{1}{MN} \sum_{m=j+1}^M \sum_{n=k+1}^N \sup_B \|Y_{m,n}(b)\| \sup_B \left\| \frac{\partial}{\partial b}[Y_{m-j,n-k}(b)] \right\|. \quad (\text{A.13}) \end{aligned}$$

The sample cross moments on the right-hand side of Eq. (A.13) can be bounded by the sample second moments:

$$\begin{aligned} & \frac{1}{MN} \sum_{m=j+1}^M \sum_{n=k+1}^N \sup_B \|Y_{m-j,n-k}(b)\| \sup_B \left\| \frac{\partial}{\partial b}[Y_{m,n}(b)] \right\| \\ & \leq \left\{ \frac{1}{MN} \sum_{m=j+1}^M \sum_{n=k+1}^N \sup_B [Y_{m-j,n-k}(b)]^2 \right\}^{1/2} \\ & \quad \times \left\{ \frac{1}{MN} \sum_{m=j+1}^M \sum_{n=k+1}^N \sup_B \left\| \frac{\partial}{\partial b}[Y_{m,n}(b)] \right\|^2 \right\}^{1/2}. \quad (\text{A.14}) \end{aligned}$$

Using Eq. (A.14) to bound the sup over j, k of the left-hand side of Eq. (A.13):

$$\begin{aligned} & \sup_{j, k} \left\| \frac{1}{MN} \sum_{m=j+1}^M \sum_{n=k+1}^N \frac{\partial}{\partial b} [Y_{m,n}(\bar{b}) Y_{m-j, n-k}(\bar{b})] \right\| \\ & \leq 2 \left\{ \frac{1}{MN} \sum_{m=1}^M \sum_{n=1}^N \sup_B [Y_{m,n}(b)]^2 \right\}^{1/2} \\ & \quad \times \left\{ \frac{1}{MN} \sum_{m=1}^M \sum_{n=1}^N \sup_B \left\| \frac{\partial}{\partial b} [Y_{m,n}(b)] \right\|^2 \right\}^{1/2} \end{aligned}$$

The assumptions $E \sup_B [Y_{m,n}(b)]^2 < \infty$ and $E \sup_B \|(\partial/\partial b)[Y_{m,n}(b)]\|^2 < \infty$ together with Markov's inequality imply

$$\begin{aligned} & 2 \left\{ \frac{1}{MN} \sum_{m=1}^M \sum_{n=1}^N \sup_B [Y_{m,n}(b)]^2 \right\}^{1/2} \\ & \quad \times \left\{ \frac{1}{MN} \sum_{m=1}^M \sum_{n=1}^N \sup_B \left\| \frac{\partial}{\partial b} [Y_{m,n}(b)] \right\|^2 \right\}^{1/2} = O_p(1). \end{aligned}$$

Therefore since $\sqrt{MN}(b_\tau - \beta)$ has an asymptotic distribution and the weights $K_{MN}(j, k)$ are uniformly bounded:

$$\frac{\sqrt{MN}}{L_M L_N} (\Gamma_2(b_\tau) - \Gamma_2(\beta)) = O_p(1)$$

Similarly for $\Gamma_2(\beta)$:

$$\begin{aligned} \frac{\sqrt{MN}}{L_M L_N} (\Gamma_2(b_\tau) - \Gamma_2(\beta)) &= \frac{1}{L_M L_N} \frac{\partial}{\partial b} \Gamma_2(\bar{b}) \sqrt{MN}(b_\tau - \beta) \\ &= \sqrt{MN}(b_\tau - \beta) \frac{1}{MN} \sum_{m=1}^M \sum_{n=1}^N \frac{\partial}{\partial b} [Y_{m,n}(\bar{b})^2]. \end{aligned}$$

Using the same argument as for Γ_1 , to bound $\sup_{j, k} \|(1/MN) \sum_{m=1}^M \sum_{n=1}^N (\partial/\partial b)[Y_{m,n}(\bar{b})^2]\|$:

$$\frac{\sqrt{MN}}{L_M L_N} (\Gamma_2(b_\tau) - \Gamma_2(\beta)) = O_p(1).$$

Therefore, since $L_M = o(M^{1/3})$ and $L_N = o(N^{1/3})$, $\Gamma(b_\tau) - \Gamma(\beta) \rightarrow 0$ in probability the proposition follows.

Proof of Proposition 4. The strategy will be to use the decomposition of partial sums in Eq. (4.1). First, we show that $Y_p(b)$ is stationary and mixing so we can apply Propositions 1 and 2 to show consistency of b_τ and obtain its limiting distribution. Then we show that the boundary terms $\sum_{s_i \in A_\tau - A^*} g(X_{s_i}; b)$ converge to zero when suitably scaled to complete the proof.

The constructed $Y_p(b)$ process can be written as product of jointly mixing, stationary processes and so inherits these properties. Write $Y_p(b) = W_p g(\tilde{X}_p; b)$. The W_p process is stationary and mixing by conditions (D5) and (D6). Stationarity and mixing of $\tilde{X}_p$ and hence $g(\tilde{X}_p; b)$ is established below.

The stationarity of the $\tilde{X}_p$ process can be seen by considering the expectation of an arbitrary function, f , of $\{\tilde{X}_{p_1}, \tilde{X}_{p_2}\}$ expressed as a conditional expectation on W_p :

$$\begin{aligned} E f(\tilde{X}_{p_1}, \tilde{X}_{p_2}) &= E \{f(X_{s_1}, X_{s_2}) | W_{p_1} = W_{p_2} = 1\} \Pr\{W_{p_1} = W_{p_2} = 1\} \\ &\quad + E \{f(X(p_1), X_{s_2}) | W_{p_1} = 0, W_{p_2} = 1\} \Pr\{W_{p_1} = 0, W_{p_2} = 1\} \\ &\quad + E \{f(X_{s_1}, X(p_2)) | W_{p_1} = 1, W_{p_2} = 0\} \Pr\{W_{p_1} = 1, W_{p_2} = 0\} \\ &\quad + E f(X(p_2), X(p_2)) \Pr\{W_{p_1} = W_{p_2} = 0\} \end{aligned} \quad (\text{A.15})$$

Stationarity of the $\tilde{X}_p$ process requires invariance of the left-hand side of (A.15) to shifts t that move the indices $\{p_1, p_2\}$ around on the set of integer coordinates.

Consider the first term on the right-hand side of equation (A.15):

$$E \{f(X_{s_1}, X_{s_2}) | W_{p_1} = W_{p_2} = 1\} \Pr\{W_{p_1} = W_{p_2} = 1\} \quad (\text{A.16})$$

Rewriting the first half of this expression:

$$E \{f(X_{s_1}, X_{s_2}) | W_{p_1} = W_{p_2} = 1\} = \int_{A_1} \int_{A_2} E \{f(X(u), X(v)) | u, v\} \tilde{\phi}(du, dv).$$

Both the functions being integrated in this expression are invariant to shifts in indexes p_1, p_2 and corresponding shifts in the sets A_1 and A_2 . $E \{f(X(u), X(v)) | u, v\}$ is invariant to shifts in index because the continuous-index random field $X(s)$ is stationary. The stationarity of the point process Φ implies that the measure on pairwise point locations $\tilde{\phi}(u, v)$ is invariant to shifts. The

remaining term in (A.16): $\Pr\{W_{p_1} = W_{p_2} = 1\}$ is invariant to shifts because the point process Φ is stationary. Thus:

$$\begin{aligned} E\{f(X_{s_1}, X_{s_2}) | W_{p_1+t} = W_{p_2+t} = 1\} \Pr\{W_{p_1+t} = W_{p_2+t} = 1\} \\ = E\{f(X_{s_1}, X_{s_2}) | W_{p_1} = W_{p_2} = 1\} \Pr\{W_{p_1} = W_{p_2} = 1\}. \end{aligned}$$

The second term in Eq. (A.15) can be written as

$$\begin{aligned} E\{f(X(p_1), X(s_2)) | W_{p_1} = 0, W_{p_2} = 1\} \Pr\{W_{p_1} = 0, W_{p_2} = 1\} \\ = \left[\int_{A_2} E\{f(X(p_1), X(v))\} \zeta(dv) \right] \Pr\{W_{p_1} = 0, W_{p_2} = 1\} \end{aligned}$$

with the measure $\zeta(v)$ denoting the measure on the locations v conditional on $\{W_{p_1} = 0, W_{p_2} = 1\}$. Again, the integrands in the bracketed term of this expression and the $\Pr\{W_{p_1} = 0, W_{p_2} = 1\}$ are invariant to translation due to the stationarity of the process $X(s)$ and the point process Φ . Invariance of the remaining terms in Eq. (A.15) to translation follows in an analogous manner. Therefore, the right-hand side of Eq. (A.15) is invariant to shifts implying

$$Ef(\tilde{X}_{p_1}, \tilde{X}_{p_2}) = Ef(\tilde{X}_{p_1+t}, \tilde{X}_{p_2+t}).$$

This same argument can be applied to a function of $\{\tilde{X}_{p_1}, \dots, \tilde{X}_{p_m}\}$ for an arbitrary m to show

$$Ef(\tilde{X}_{p_1}, \dots, \tilde{X}_{p_m}) = Ef(\tilde{X}_{p_1+t}, \dots, \tilde{X}_{p_m+t})$$

implying that the process $\tilde{X}_p$ is stationary.

The assumption that the continuous-index random field is mixing implies that $\tilde{X}_p$ is mixing as well. Getting mixing conditions for the constructed lattice process involves a change in metrics from Euclidean to maximum coordinate-wise distance as above and keeping track of the normalization that $d^* = 1$. Because of the $d^* = 1$ normalization, the Euclidean distance mixing coefficient for sets n units apart equals $\alpha_{k,l}$ for d^*n in the original units. Tail properties of this sequence are unchanged and B1–B3 will hold for ‘ $d^* = 1$ ’ units.

Changing to the coordinatewise metric presents another technical detail. The definition of the latent process $\tilde{X}_p$ to be equal to the value of the continuous random field at the point p if $W_p = 0$ means that Euclidean distances between the indices of the continuous indexed random fields that are relabelled $\tilde{X}_p$ and $\tilde{X}_{p'}$ could be one unit (length of the side of a square) less than the distance between p and p' . For example, the indices of the continuous indexed random

fields that are relabelled $\tilde{X}_p$ and $\tilde{X}_{p+[0,1]}$ could be arbitrarily small if $\tilde{X}_p$ were equal to the continuous random field X_s at $s = p$ (because $W_p = 0$) and $\tilde{X}_{p+[0,1]}$ were equal to $X_{p+\epsilon}$ for an arbitrarily small positive vector ϵ . Therefore, $\tilde{\alpha}_{k,l}(1)$ will be one and the sequence of mixing coefficients for greater distances will be shifted one place for $n \geq 2$: $\tilde{\alpha}_{k,l}(n) = \alpha_{k,l}(n-1)$. The mixing conditions B1', B2', B3' will still hold for the $\tilde{X}_p$ process because the tail properties of the $\tilde{\alpha}_{k,l}(n)$ sequence are unchanged by this shift.

Consistency and asymptotic normality. A uniform law of large numbers can be obtained for $(1/T_\tau) \sum_{p \in A_\tau^*} Y_p(b)$ using the same argument as in Proposition 1. Write

$$\frac{1}{T_\tau} \sum_{p \in A_\tau^*} Y_p(b) = \left[\frac{|A_\tau^*|}{T_\tau} \right] \left[\frac{1}{|A_\tau^*|} \sum_{p \in A_\tau^*} Y_p(b) \right]. \quad (\text{A.17})$$

As in the proof of Proposition 1 an ergodic theorem can be applied to each of the bracketed terms in (A.17). The ergodic theorem applied to the first term yields

$$\frac{1}{|A_\tau^*|} \sum_{p \in A_\tau^*} Y_p(b) \rightarrow \mathbb{E} Y_p(b) = \lambda \text{Eg}(X_{s_p}; b), \text{ w.p.1 as } \tau \rightarrow \infty$$

Next consider the term $|A_\tau^*|/T_\tau$ in Eq. (A.17), writing the inverse of this term in terms of W_p :

$$\frac{T_\tau}{|A_\tau^*|} = \frac{1}{|A_\tau^*|} \sum_{p \in A_\tau^*} W_p + \frac{1}{|A_\tau^*|} \sum_{s_i \in A_\tau} 1(s_i \in A_\tau - A_\tau^*),$$

where $1(\cdot)$ is an indicator function. The ergodic theorem can be applied as above to the first term to imply that $(1/|A_\tau^*|) \sum_{p \in A_\tau^*} W_p \rightarrow \mathbb{E} W_p$, w.p.1. The term $(1/|A_\tau^*|) \sum_{s_i \in A_\tau} 1(s_i \in A_\tau - A_\tau^*) \rightarrow 0$ because the largest the number of terms in $\sum_{s_i \in A_\tau} 1(s_i \in A_\tau - A_\tau^*)$ can be is $O(\sqrt{|A_\tau^*|})$:

$$\frac{T_\tau}{|A_\tau^*|} \rightarrow \lambda \text{ w.p.1. as } \tau \rightarrow \infty. \quad (\text{A.18})$$

Slutsky's theorem then implies

$$\frac{1}{T_\tau} \sum_{p \in A_\tau^*} Y_p(b) \rightarrow \text{Eg}(X_{s_p}; b) \text{ w.p.1. as } \tau \rightarrow \infty.$$

Now, the remainder term can be written as

$$\frac{1}{T_\tau} \sum_{s_i \in A_\tau - A_\tau^*} g(X_{s_i}; b) = \frac{1}{\sqrt{|A_\tau^*|}} \left[\frac{|A_\tau^*|}{T_\tau} \right] \left[\frac{1}{\sqrt{|A_\tau^*|}} \sum_{s_i \in A_\tau - A_\tau^*} g(X_{s_i}; b) \right].$$

The ergodic theorem can be applied as above to each of the bracketed terms to show that they converge. The remaining factor of $(1/\sqrt{|A_\tau^*|})$ implies that $(1/T_\tau) \sum_{s_i \in A_\tau - A_\tau^*} g(X_{s_i}; b) \rightarrow 0$ w.p.1. Therefore, we get a pointwise LLN:

$$\frac{1}{T_\tau} \sum_{i=1}^{T_\tau} g(X_{s_i}; b) \rightarrow \text{E}g(X_{s_i}; b) \text{ w.p.1. as } \tau \rightarrow \infty. \quad (\text{A.19})$$

The remainder of a consistency proof is the same as that for Proposition 1.

As previewed above, a central limit result for $(1/\sqrt{T_\tau}) \sum_{i=1}^{T_\tau} g(X_{s_i}; \beta)$ is obtained by approximating this sum of the constructed integer lattice random field plus the boundary points:

$$\begin{aligned} \frac{1}{\sqrt{T_\tau}} \sum_{i=1}^{T_\tau} g(X_{s_i}; \beta) &= \frac{\sqrt{|A_\tau^*|}}{\sqrt{T_\tau}} \frac{1}{\sqrt{|A_\tau^*|}} \sum_{p \in A_\tau^*} Y_p(\beta) \\ &\quad + \frac{\sqrt{|A_\tau^*|}}{\sqrt{T_\tau}} \frac{1}{\sqrt{|A_\tau^*|}} \sum_{s_i \in A_\tau - A_\tau^*} g(X_{s_i}; \beta). \end{aligned} \quad (\text{A.20})$$

Slutsky's theorem and Eq. (A.18) imply $\sqrt{|A_\tau^*|}/\sqrt{T_\tau} \rightarrow \lambda^{-1/2}$. Since $Y_p(\beta)$ is stationary and mixing satisfying B1'–B3'; and conditions A1, B4 and B5 hold; Bolthausen (1982) can be directly applied as in Proposition 2 to obtain a limiting distribution for the first term on the right-hand side of Eq. (A.20):

$$\frac{\sqrt{|A_\tau^*|}}{\sqrt{T_\tau}} \frac{1}{\sqrt{|A_\tau^*|}} \sum_{p \in A_\tau^*} Y_p(\beta) \Rightarrow N(0, \lambda^{-1} V).$$

The number of elements in the second term in Eq. (A.20) depends on the exact sequence of sample regions A_τ . However, the largest number of terms it can have is $O(\sqrt{|A_\tau^*|})$. In this case, the Ergodic theorem can be applied to show that $(1/\sqrt{|A_\tau^*|}) \sum_{s_i \in A_\tau - A_\tau^*} g(X_{s_i}; \beta) \rightarrow c \text{E}g(X_{s_i}; \beta)$ w.p.1, for some $c < \infty$ and since $\text{E}g(X_{s_i}; \beta) = 0$:

$$\frac{1}{\sqrt{|A_\tau^*|}} \sum_{s_i \in A_\tau - A_\tau^*} g(X_{s_i}; \beta) \rightarrow 0 \text{ in prob. as } \tau \rightarrow \infty.$$

For sequences A_τ resulting in a number of terms that is of a smaller order than $\sqrt{|A_\tau^*|}$, $1/\sqrt{|A_\tau^*|} \sum_{s_i \in A_\tau - A_\tau^*} g(X_{s_i}; \beta)$ will converge even more quickly to zero. Thus, the boundary terms do not contribute to the limiting distribution of $(1/\sqrt{T_\tau}) \sum_{i=1}^{T_\tau} g(X_{s_i}; \beta)$. A uniform law of large numbers can be constructed as above for each of the partial sums of derivative terms in expression (3.7) to show they converge in probability to D_0 . The proposition then follows via Slutsky's theorem.

Proof of Proposition 5. The strategy is essentially the same as in the proof of Proposition 3. Therefore we will only highlight the differences in key equations. To facilitate indexing of points, $\tilde{C}_\tau$ can be expressed as a weighted average of sample points using the same method to index points as in Proposition 3, because each observation $g(X_{s_i}; b_\tau)$ was given a unique index (m, n) . However, now the weights that each product of observations receives will be random variables that depend on the measurement errors and on the choice of $\tilde{d}$ as well as the chosen window $K(\cdot, \cdot)$ and cutoff parameters L_M and L_N . Thus,

$$\begin{aligned} \tilde{C}_\tau = & \frac{1}{T_\tau} \sum_{j=0}^{L_M} \sum_{k=0}^{L_N} \sum_{m=j+1}^M \sum_{n=k+1}^N \tilde{K}_{MN}(m, n, j, k) \begin{pmatrix} Y_{m,n}(b_\tau) Y_{m-j, n-k}(b_\tau)' + \\ Y_{m-j, n-k}(b_\tau) Y_{m,n}(b_\tau)' \end{pmatrix} \\ & - \frac{1}{T_\tau} \sum_{m=1}^M \sum_{n=1}^N Y_{m,n}(b_\tau) Y_{m,n}(b_\tau)', \end{aligned}$$

where $\tilde{L}_M \equiv LI[\tilde{d}/d^*](L_M + 1)$ and $\tilde{L}_N \equiv LI[\tilde{d}/d^*](L_N + 1)$ with $LI[\tilde{d}/d^*]$ denoting the least integer greater than $\tilde{d}/d^*$. The random function $\tilde{K}_{MN}(m, n, j, k)$ is a function of random displacements caused by measurement errors and deterministic displacements resulting from the use of $\tilde{d}$ rather than d^* to define squares. We will write the distance in each direction between points (m, n) and $(m-j, n-k)$ as having three components. For example, in the M direction $GI[d^*/\tilde{d}j]$ will capture the majority of the deterministic component of distance and u_{mn}^M will denote a random shift that is due to measurement error plus a rounding error for the (m, n) point and $u_{m-j, n-k}^M$ is the counterpart for the $(m-j, n-k)$ point. Thus, letting u_{mn}^M be the displacement of the (m, n) point in the M direction from measurement and rounding error and u_{mn}^N be its counterpart in the N direction (suppressing dependence on $\tilde{d}$), write the random weight:

$$\tilde{K}_{MN}(m, n, j, k) = K \left(\frac{GI[\frac{d^*}{\tilde{d}j}] + u_{mn}^M - u_{m-j, n-k}^M}{L_M}, \frac{GI[\frac{d^*}{\tilde{d}k}] + u_{mn}^N - u_{m-j, n-k}^N}{L_N} \right).$$

While the distribution of the displacements, $\{u_{mn}^M, u_{mn}^N\}$ is unknown, its support can be derived using the bound on the measurement error process. Letting the bound on the support of measurement errors be $|e| \leq A$, then the support of

$\{u_{mn}^M, u_{mn}^N\}$ for all m, n is contained in the set of integers in the interval $[\pm(LI[A/\tilde{d}] + \{2LI[d^*/\tilde{d}] + 1\})]$. The term $LI[A/\tilde{d}]$ reflects the maximum displacement due to measurement error and the remaining term captures maximum contribution of rounding.

As in the proof of Proposition 3 we take the case of scalar $Y_{m,n}(\cdot)$ and first prove consistency of the analog of $\hat{V}$. Proceeding as in the proof of Proposition 3, define an array:

$$Z_{MN, mn} = \sum_{j=0}^{\tilde{L}_M} \sum_{k=0}^{\tilde{L}_N} \left(\tilde{K}_{MN}(m, n, j, k) Y_{m,n}(\beta) Y_{m-j, n-k}(\beta) - E \tilde{K}_{MN}(m, n, j, k) Y_{m,n}(\beta) Y_{m-j, n-k}(\beta) \right).$$

Lemma A follows with $\tilde{L}_M, \tilde{L}_N$ replacing L_M and L_N throughout and the following modifications in its proof. The analog to the right-hand side of Eq. (A.5) can be divided into close terms and far terms with close defined as within $2\tilde{L}_M$ and $2\tilde{L}_N$ and the remainder as far.

The bound on $\sup_{1 \leq m' \leq M, 1 \leq n' \leq N} \|Z_{MN, m'n'}\|_2^2$ used to bound the sum of the close terms is altered. Using Minkowski's inequality we arrive at the following bound instead of Eq. (A.6):

$$\|Z_{MN, mn}\|_q \leq \sum_{j=0}^{\tilde{L}_M} \sum_{k=0}^{\tilde{L}_N} \left\| \tilde{K}_{MN}(m, n, j, k) Y_{m,n}(\beta) Y_{m-j, n-k}(\beta) - E \tilde{K}_{MN}(m, n, j, k) Y_{m,n}(\beta) Y_{m-j, n-k}(\beta) \right\|_q.$$

The random weight $\tilde{K}_{MN}(m, n, j, k)$ has uniformly bounded support and $Y_{m,n}(\beta)$ has $(4 + \delta)$ th moments so

$$\left\| \tilde{K}_{MN}(m, n, j, k) Y_{m,n}(\beta) Y_{m-j, n-k}(\beta) - E \tilde{K}_{MN}(m, n, j, k, \tilde{d}) Y_{m,n}(\beta) Y_{m-j, n-k}(\beta) \right\|_q < \infty$$

and

$$\|Z_{MN, mn}\|_q \leq c_5(\tilde{L}_M + 1)(\tilde{L}_N + 1)$$

for $q = 2, 2 + \delta$. Thus, the bounds in Eq. (A.7) obtain for this definition of $Z_{MN, mn}$ with $\tilde{L}_M, \tilde{L}_N$ replacing L_M, L_N .

Condition E1 implies that $Z_{MN, mn}$ is mixing and therefore Inequality A holds for this altered definition of $Z_{MN, mn}$. This inequality can be used to obtain a bound on the contribution of the far terms exactly as in the proof of Proposition 3. The maximum number of far and near terms are also the same as for Proposition 3 with $\tilde{L}_M, \tilde{L}_N$ replacing L_M, L_N . Thus, the analog to Eq. (A.10)

takes the following form:

$$\begin{aligned} \text{var} \left\{ \sum_{m=1}^M \sum_{n=1}^N Z_{MN,mn} \right\} &\leq c_6 \{ MN(4\tilde{L}_M + 1)(4\tilde{L}_N + 1)(\tilde{L}_M + 1)^2(\tilde{L}_N + 1)^2 \\ &+ c_7(MN)^2 \tilde{\alpha}_{\infty, \infty}(\min(\tilde{L}_M, \tilde{L}_N))^{\delta/(2+\delta)}(\tilde{L}_M + 1)^2(\tilde{L}_N + 1)^2 \} \end{aligned} \quad (\text{A.21})$$

for some constants c_6 and c_7 . The assumption that $L_M = o(M^{1/3})$ and $L_N = o(N^{1/3})$ implies $\tilde{L}_M = o(N^{1/3})$ and $\tilde{L}_N = o(M^{1/3})$ and $\tilde{\alpha}_{\infty, \infty}(m)^{\delta/(2+\delta)} = o(m^{-4})$ so Lemma A follows for this $Z_{MN,mn}$ using Chebychev's inequality as in the proof of Proposition 3.

The next step, analogous to showing that $E\hat{V}$ gets close to V , is altered. The analog of the expression for $E\hat{V} - \text{var}\{(1/\sqrt{MN})\sum_{m=1}^M \sum_{n=1}^N Y_{m,n}(\beta)\}$ in Proposition 3 is

$$\begin{aligned} &\frac{2}{MN} \sum_{j=0}^{\tilde{L}_M} \sum_{k=0}^{\tilde{L}_N} \sum_{m=j+1}^M \sum_{n=k+1}^N E\{(\tilde{K}_{MN}(m, n, j, k) - 1)Y_{m,n}(\beta)Y_{m-j, n-k}(\beta)\} \\ &- \frac{1}{MN} \sum_{j=\tilde{L}_M+1}^{M-1} \sum_{k=\tilde{L}_N+1}^{N-1} \sum_{m=j+1}^M \sum_{n=k+1}^N EY_{m,n}(\beta)Y_{m-j, n-k}(\beta). \end{aligned} \quad (\text{A.22})$$

To show that the first term converges to zero we bound $|E\{(\tilde{K}_{MN}(m, n, j, k) - 1)Y_{m,n}(\beta)Y_{m-j, n-k}(\beta)\}|$ by terms whose sum converges, demonstrate pointwise convergence of $E\{(\tilde{K}_{MN}(m, n, j, k) - 1)Y_{m,n}(\beta)Y_{m-j, n-k}(\beta)\}$ to zero and apply the dominated convergence theorem.

$$\begin{aligned} &|E\{(\tilde{K}_{MN}(m, n, j, k) - 1)Y_{m,n}(\beta)Y_{m-j, n-k}(\beta)\}| \\ &\leq |E\{\tilde{K}_{MN}(m, n, j, k)Y_{m,n}(\beta)Y_{m-j, n-k}(\beta)\}| + |E\{Y_{m,n}(\beta)Y_{m-j, n-k}(\beta)\}| \end{aligned} \quad (\text{A.23})$$

To get a bound on $|E\{\tilde{K}_{MN}(m, n, j, k)Y_{m,n}(\beta)Y_{m-j, n-k}(\beta)\}|$ we write $\tilde{K}_{MN}(m, n, j, k)$ as a sum of products of functions of $u_{m,n}^M$, $u_{m,n}^N$ and $u_{m-j, n-k}^M$, $u_{m-j, n-k}^N$ using the Fourier series representation for $K(\cdot, \cdot)$:

$$K(l, v) = \sum_{t=-\infty}^{\infty} \sum_{r=-\infty}^{\infty} A_{t,r} \exp(i\pi(tl + rv)).$$

We write $\tilde{K}_{MN}(m, n, j, k)Y_{m,n}(\beta)Y_{m-j,n-k}(\beta)$ as an infinite sum with terms that are products of functions of u 's times $Y_{m,n}(\beta)$ and $Y_{m-j,n-k}(\beta)$:

$$\begin{aligned}\tilde{K}_{MN}(m, n, j, k)Y_{m,n}(\beta)Y_{m-j,n-k}(\beta) &= \sum_{t=-\infty}^{\infty} \sum_{r=-\infty}^{\infty} A_{t,r} \exp\left(i\pi t \frac{GI \left[\frac{d^*}{d}j\right]}{L_M}\right) \\ &\exp\left(i\pi t \frac{u_{mn}^M}{L_M}\right) \exp\left(-i\pi t \frac{u_{m-j,n-k}^M}{L_M}\right) \exp\left(i\pi r \frac{GI \left[\frac{d^*}{d}k\right]}{L_N}\right) \\ &\exp\left(i\pi r \frac{u_{mn}^N}{L_N}\right) \exp\left(-i\pi r \frac{u_{m-j,n-k}^N}{L_N}\right) Y_{m,n}(\beta) Y_{m-j,n-k}(\beta).\end{aligned}\quad (\text{A.24})$$

Terms can then be grouped into products of variables with common indices. Each term in the sum on the right-hand side of (A.24) can be written as

$$\begin{aligned}&\left[A_{t,r} \exp\left(i\pi t \frac{GI \left[\frac{d^*}{d}j\right]}{L_M}\right) \exp\left(i\pi r \frac{GI \left[\frac{d^*}{d}k\right]}{L_M}\right) \right] \\ &\left[\exp\left(i\pi t \frac{u_{mn}^M}{L_M}\right) \exp\left(i\pi r \frac{u_{mn}^N}{L_N}\right) Y_{m,n}(\beta) \right] \\ &\left[\exp\left(-i\pi t \frac{u_{m-j,n-k}^M}{L_M}\right) \exp\left(-i\pi r \frac{u_{m-j,n-k}^N}{L_N}\right) Y_{m-j,n-k}(\beta) \right] \\ &= A_{t,r} \psi_{t,r}(j, k) F_{mn} F_{m-j,n-k}.\end{aligned}$$

Now,

$$E\tilde{K}_{MN}(m, n, j, k)Y_{m,n}(\beta)Y_{m-j,n-k}(\beta) = \sum_{t=-\infty}^{\infty} \sum_{r=-\infty}^{\infty} A_{t,r} \psi_{t,r}(j, k) E F_{mn} F_{m-j,n-k}$$

and

$$\begin{aligned}&\sum_{t=-\infty}^{\infty} \sum_{r=-\infty}^{\infty} A_{t,r} \psi_{t,r}(j, k) E F_{mn} F_{m-j,n-k} \\ &= \sum_{t=-\infty}^{\infty} \sum_{r=-\infty}^{\infty} A_{t,r} \psi_{t,r}(j, k) \text{cov}(F_{mn}, F_{m-j,n-k}) \\ &+ \sum_{t=-\infty}^{\infty} \sum_{r=-\infty}^{\infty} A_{t,r} \psi_{t,r}(j, k) E F_{mn} E F_{m-j,n-k}.\end{aligned}$$

Using Inequality A,

$$|cov(F_{mn}, F_{m-j, n-k})| \leq c_8 \tilde{\alpha}_{\infty, \infty} (\min(j, k))^{\delta/(2+\delta)}$$

for some constant c_8 . Since the Fourier coefficients are absolutely summable:

$$\begin{aligned} & \left| \sum_{t=-\infty}^{\infty} \sum_{r=-\infty}^{\infty} A_{t,r} \psi_{t,r}(j, k) cov(F_{mn}, F_{m-j, n-k}) \right| \\ & \leq \sum_{t=-\infty}^{\infty} \sum_{r=-\infty}^{\infty} |A_{t,r} \psi_{t,r}(j, k)| |cov(F_{mn}, F_{m-j, n-k})| \\ & \leq c_9 \tilde{\alpha}_{\infty, \infty} (\min(j, k))^{\delta/(2+\delta)}. \end{aligned}$$

$EF_{mn}EF_{m-j, n-k}$ can be bounded using the following argument:

$$EF_{mn} = E \exp\left(i\pi t \frac{u_{mn}^M}{L_M}\right) \exp\left(i\pi r \frac{u_{mn}^N}{L_N}\right) Y_{m,n}(\beta).$$

Using the fact that the support of u_{mn}^M and u_{mn}^N is finite write EF_{mn} as

$$\begin{aligned} & \sum_{U_1} \sum_{U_2} \exp\left(i\pi t \frac{U_1}{L_M}\right) \exp\left(i\pi r \frac{U_2}{L_N}\right) E(Y_{m,n}(\beta) | u_{mn}^M \\ & = U_1, u_{mn}^N = U_2) \Pr(u_{mn}^M = U_1, u_{mn}^N = U_2); \end{aligned}$$

of course

$$EY_{m,n}(\beta) = \sum_{U_1} \sum_{U_2} E(Y_{m,n}(\beta) | u_{mn}^M = U_1, u_{mn}^N = U_2) \Pr(u_{mn}^M = U_1, u_{mn}^N = U_2).$$

Therefore, since condition E2 implies $E(Y_{m,n}(\beta) | u_{mn}^M = U_1, u_{mn}^N = U_2) < \infty$ for all of the finite number of points of support for the u 's:

$$\begin{aligned} EF_{mn} - EY_{m,n}(\beta) &= O\left[\exp\left(i\pi t \frac{\bar{U}}{L_M}\right) \exp\left(i\pi r \frac{\bar{U}}{L_N}\right) - 1\right] \\ &= O\left(\max\left(\frac{1}{L_M}, \frac{1}{L_N}\right)\right), \end{aligned}$$

where $\bar{U}$ is the maximum of the support of the u 's. Since $EY_{m,n}(\beta) = 0$, $EF_{mn} = O(1/L_M)$ (L_M and L_N are of the same order). The same argument can be used on the other term to imply $EY_{m,n}(\beta)EY_{m-j,n-k}(\beta) = O(1/L_M^2)$. Thus

$$\left| \sum_{t=-\infty}^{\infty} \sum_{r=-\infty}^{\infty} A_{t,r} \psi_{t,r}(j,k) EF_{mn} EF_{m-j,n-k} \right| \leq c_{10}(1/L_M^2)$$

and therefore

$$|E\tilde{K}_{MN}(m,n,j,k)Y_{m,n}(\beta)Y_{m-j,n-k}(\beta)| \leq c_9 \tilde{\alpha}_{\infty,\infty}(\min(j,k))^{\delta/(2+\delta)} + c_{10}(1/L_M^2).$$

Inequality A can be applied to the second term in Eq. (A.23) to show that

$$|EY_{m,n}(\beta)Y_{m-j,n-k}(\beta)| \leq c_{11} \tilde{\alpha}_{\infty,\infty}(\min(j,k))^{\delta/(2+\delta)}.$$

Therefore, the sum of the left-hand side of Eq. (A.23) can be bounded as

$$\begin{aligned} & \frac{1}{MN} \sum_{j=0}^{L_M} \sum_{k=0}^{L_N} \sum_{m=j+1}^M \sum_{n=k+1}^N |E\{(\tilde{K}_{MN}(m,n,j,k) - 1)Y_{m,n}(\beta)Y_{m-j,n-k}(\beta)\}| \\ & \leq \frac{1}{MN} \sum_{j=0}^{L_M} \sum_{k=0}^{L_N} \sum_{m=j+1}^M \sum_{n=k+1}^N \{(c_9 + c_{11})\tilde{\alpha}_{\infty,\infty} \\ & (\min(j,k))^{\delta/(2+\delta)} + c_{10}(1/L_M^2)\} < \infty, \end{aligned}$$

where the right-hand side is finite since $\tilde{\alpha}_{\infty,\infty}(m)^{\delta/(2+\delta)} = o(m^{-4})$. All the points of support for $\tilde{K}$ converge to 1 as $M, N \rightarrow \infty$ for each j, k so

$$E(\tilde{K}_{MN}(m,n,j,k) - 1)Y_{m,n}(\beta)Y_{m-j,n-k}(\beta) \rightarrow 0 \text{ w.p.1.}$$

Therefore, the dominated convergence theorem implies the first term in the expression (A.22) converges to zero. Inequality A can be used on the term

$$\sum_{j=L_M+1}^{M-1} \sum_{k=L_N+1}^{N-1} \sum_{m=j+1}^M \sum_{n=k+1}^N EY_{m,n}(\beta)Y_{m-j,n-k}(\beta)$$

as in Proposition 3 to bound it by the tail of convergent double sequence, completing the proof of consistency for the infeasible version of $\tilde{C}$ estimator that uses β instead of b_{τ} .

The introduction of bounded measurement errors does not impact the remainder of the proof beyond rewriting terms to include $\tilde{K}_{MN}(m, n, j, k)$. Bounds on the norms of derivatives will be unchanged, e.g.

$$\left\| \tilde{K}_{MN}(m, n, j, k) \frac{\partial}{\partial b} [Y_{m,n}(\bar{b})] Y_{m-j, n-k}(\bar{b}) \right\| \\ \leq \sup \| \tilde{K}_{MN}(m, n, j, k) \| \sup_B \| Y_{m-j, n-k}(b) \| \sup_B \left\| \frac{\partial}{\partial b} [Y_{m,n}(b)] \right\|.$$

$\tilde{K}_{MN}(m, n, j, k)$ is uniformly bounded so the argument is the same as in Proposition 3.

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